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Acoustics - Determination of occupational noise exposure - Methodology (ISO 9612:2025,IDT)

Vervangt NEN-EN-ISO 9612:2009

ICS 13.140

juni 2025

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EUROPEAN STANDARD

NORME EUROPÉENNE

EUROPÄISCHE NORM

EN ISO 9612

June 2025

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Supersedes EN ISO 9612:2009

English Version

Acoustics - Determination of occupational noise exposure - Methodology (ISO 9612:2025)

Acoustique - Détermination de l'exposition au bruit en milieu de travail - Méthodologie (ISO 9612:2025)

Akustik - Bestimmung der Lärmexposition am Arbeitsplatz - Methodik (ISO 9612:2025)

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Contents

Page

European foreword **3**

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International Standard

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Acoustics — Determination of occupational noise exposure — Methodology

Acoustique — Détermination de l'exposition au bruit en milieu de travail — Méthodologie

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Contents

Page

Foreword ..... **v**

Introduction ..... **vi**

1 Scope **1**

2 Normative references **1**

3 Terms and definitions **1**

4 Symbols **3**

5 Instrumentation **5**
5.1 Sound level meters and personal sound exposure meters 5
5.2 Calibrator 5
5.3 Periodic verification 6

6 Methodology – Chronological steps **6**
6.1 Step 1: Work analysis 6
6.2 Step 2: Selection of the measurement strategy 6
6.3 Step 3: Measurements 6
6.4 Step 4: Error handling and uncertainties 6
6.5 Step 5: Calculation and presentation of results and uncertainty 6

7 Work analysis **6**
7.1 General 6
7.2 Defining homogeneous noise exposure groups (HEG) 7
7.3 Determination of a nominal day 7

8 Selection of measurement strategies **8**
8.1 General 8
8.2 Measurement strategies 8

9 Strategy 1 – Task-based measurement **8**
9.1 Dividing the nominal day into tasks 8
9.2 Duration of tasks 8
9.3 Measurement of $L_{p,\mathrm{A,eq}T_m}$ for tasks 9
9.3.1 General 9
9.3.2 Number of measurements 9
9.3.3 Time and duration of measurements 10
9.3.4 Calculation of the A-weighted equivalent continuous sound pressure level 10
9.4 Calculation of contribution from each task to daily noise exposure level 10
9.5 Determination of daily noise exposure level 11

10 Strategy 2 – Job-based measurement **12**
10.1 General 12
10.2 Measurement plan – Number, duration and distribution of measurements 12
10.3 Measurements 12
10.4 Determination of daily noise exposure levels for workers in a homogeneous noise exposure group 13

11 Strategy 3 — Full-day measurement **13**
11.1 General 13
11.2 Observing work activities and monitoring measurements 14
11.3 Measurements 14
11.4 Determination of daily noise exposure level 15

12 Measurements **15**
12.1 Selection of instrumentation 15
12.2 Field calibration 15
12.3 Instrument worn by the worker 15

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12.4 Integrating-averaging sound level meter ... 16

13 Sources of uncertainty and errors ... 17

13.1 General ... 17

13.2 Mechanical impacts on microphone ... 17

13.3 Wind and airflows ... 18

13.4 Relevance of sound contributions ... 18

14 Calculation of measurement uncertainties and presentation of the final results ... 18

15 Information to be reported ... 18

Annex A (informative) **Example of a checklist to ensure that significant noise events are detected during the work analysis ... 21**

Annex B (informative) **Guidance to the selection of measurement strategy ... 22**

Annex C (normative) **Evaluation of measurement uncertainties ... 25**

Annex D (informative) **Example showing calculation of daily noise exposure level using task-based measurements ... 34**

Annex E (informative) **Example showing calculation of daily noise exposure level using job-based measurements ... 39**

Annex F (informative) **Sample calculation of daily noise exposure level using full-day measurements ... 42**

Annex G (informative) **Example calculation of daily noise exposure level for flexible workers ... 45**

Annex H (normative) **Uncertainty calculation for peak sound pressure levels ... 49**

Bibliography ... 53

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This document was prepared by Technical Committee ISO/TC 43, *Acoustics*, Subcommittee SC 1, *Noise,* in collaboration with the European Committee for Standardization (CEN) Technical Committee CEN/TC 211, *Acoustics,* in accordance with the Agreement on technical cooperation between ISO and CEN (Vienna Agreement).

This second edition cancels and replaces the first edition (ISO 9612:2009), which has been technically revised.

The main changes are as follows:

— A reshaping of the guidance on measurement of $L_{p,\mathrm{A,eq}T_m}$ for the task-based strategy (9.3).

— A revision of the measurement plan for the job-based strategy (10.2).

— The addition of Homogeneous noise Exposure Groups (HEG) sampling requirements for the full-day measurement strategy and the addition of criteria to validate sampling (11.3).

— Some precisions and clarifications on the instrumentation section.

— Some additions to the test report section: number of peak events, $L_{\mathrm{EX,8h,95\,\%}}$.

— The addition of C.7 in Annex C, which gives the formulae to calculate the measurement uncertainty when multiple nominal days are used. An Annex H is also introduced to clarify uncertainty of peak measurements.

— The introduction of a new Annex G.

— The introduction of a new Annex H.

— A full revision of the Excel calculation file attached to this document.

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Introduction

This document provides a stepwise approach to the determination of occupational noise exposure from noise level measurements. The procedure contains the following major steps: work analysis, selection of measurement strategy, measurements, error handling and uncertainty evaluations, calculations, and presentation of results. This document specifies three different measurement strategies: task-based measurement; job-based measurement; and full-day measurement. This document gives guidance on selecting an appropriate measurement strategy for a particular work situation and purpose of investigation. This document also provides an informative spreadsheet to allow calculation of measurement results and uncertainties. ISO is not responsible for errors that shall arise or occur with the use of this spreadsheet.

This document recognizes the use of hand-held sound level meters as well as personal sound exposure meters. The methods specified optimize the effort required for obtaining a given accuracy.

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Acoustics — Determination of occupational noise exposure — Methodology

1 Scope

This document specifies a method for measuring workers' exposure to noise in a working environment and calculating the noise exposure level. This document deals with A-weighted levels but is applicable also to C-weighted levels. Three different strategies for measurement are specified. The method is applicable for detailed noise exposure studies or epidemiological studies of hearing damage or other adverse effects.

The measuring process requires observation and analysis of the noise exposure conditions so that the quality of the measurements can be controlled. This document provides methods for estimating the uncertainty of the results.

This document is not intended for assessment of masking of oral communication or assessment of infrasound, ultrasound and non-auditory effects of noise. It does not apply to the measurement of the noise exposure of the ear when hearing protectors are worn.

Results of the measurements performed in accordance with this document can provide useful information when defining priorities for noise control measures.

2 Normative references

The following documents are referred to in the text in such a way that some or all of their content constitutes requirements of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

ISO 1999, *Acoustics — Estimation of noise-induced hearing loss*

ISO/IEC Guide 98-3, *Uncertainty of measurement — Part 3: Guide to the expression of uncertainty in measurement (GUM:1995)*

IEC 60942, *Electroacoustics — Sound calibrators*

IEC 61252, *Electroacoustics — Specifications for personal sound exposure meters*

IEC 61672-1, *Electroacoustics — Sound level meters — Part 1: Specifications*

3 Terms and definitions

For the purposes of this document, the following terms and definitions apply.

ISO and IEC maintain terminology databases for use in standardization at the following addresses:

— ISO Online browsing platform: available at https://www.iso.org/obp

— IEC Electropedia: available at https://www.electropedia.org/

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3.1
A-weighted equivalent continuous sound pressure level
$L_{p,\mathrm{A,eq}T}$
ten times the logarithm to the base 10 of the ratio of the time average of the square of the A-weighted sound pressure, p_A, during a stated time interval of duration T (starting at t_1 and ending at t_2), to the square of a reference value, p_0, expressed in decibels:

$$L_{p,\mathrm{A,eq}T} = 10\lg\left[\frac{\frac{1}{T}\int_{t_1}^{t_2} p_\mathrm{A}^2(t)\,\mathrm{d}t}{p_0^2}\right]\mathrm{dB}$$

where the reference sound pressure, p_0, is 20 µPa

Note 1 to entry: Adapted from ISO/TR 25417:2007[9].

3.2
A-weighted noise exposure level normalized to an 8 h work day
<daily noise exposure level>
$L_\mathrm{EX,8h}$
daily noise exposure level, in decibels, given by the formula:

$$L_\mathrm{EX,8h} = L_{p,\mathrm{A,eq}T_\mathrm{e}} + 10\lg\left[\frac{T_\mathrm{e}}{T_0}\right]\mathrm{dB}$$

where

$L_{p,\mathrm{A,eq}T_\mathrm{e}}$	is the A-weighted equivalent continuous sound pressure level for T_e;
T_e	is the effective duration, in hours, of the working day;
T_0	is the reference duration, T_0 = 8 h.

Note 1 to entry: If the effective duration of the working day, T_e, is equal to 8 h, then $L_\mathrm{EX,8h}$ equals $L_{p,\mathrm{A,eq8h}}$.

Note 2 to entry: If the average or normalized exposure over a number of days is desired, the following formula can be used:

$$\bar{L}_\mathrm{EX,8h} = 10\lg\left[\frac{1}{X}\sum_{x=1}^{X} 10^{0,1\cdot L_{\mathrm{EX,8h},x}}\right]\mathrm{dB}$$

where $L_{\mathrm{EX,8h},x}$ is the daily noise exposure level for day x.

The value of X is chosen according to the purpose of the averaging process. For example, X = 5 leads to a daily noise exposure level normalized to a nominal week of five 8 h working days.

Note 3 to entry: This definition differs from that given in ISO/TR 25417:2007[9].

3.3
nominal day
working day over which it is chosen to determine the noise exposure

Note 1 to entry: The nominal day is determined from the work analysis and depends on the purpose of the measurements. See also 7.3.

Note 2 to entry: The noise exposure level is normally calculated on a daily basis, but there can be circumstances where the use of weekly or longer noise exposure periods is considered appropriate.

Note 3 to entry: The nominal day can be a synthesized working day that includes fragments of different real working days of a week or of several consecutive weeks.

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3.4
C-weighted peak sound pressure level
$L_{p,\mathrm{Cpeak}}$
ten times the logarithm to the base 10 of the ratio of the square of the C-weighted peak sound pressure, p_{Cpeak}, to the square of a reference value, p_0, expressed in decibels

$$L_{p,\mathrm{Cpeak}} = 10 \lg \frac{p_{\mathrm{Cpeak}}^2}{p_0^2} \,\mathrm{dB}$$

where the reference sound pressure, p_0 is 20 µPa

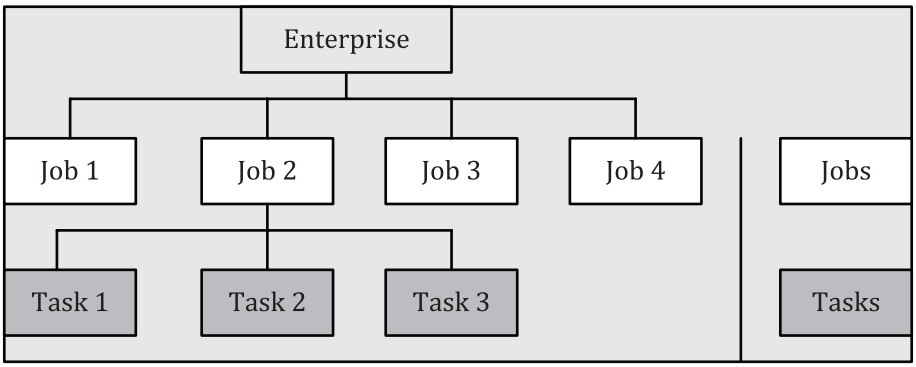
3.5
task
distinct part of a worker´s occupational activity

Note 1 to entry: Figure 1 illustrates the hierarchy of jobs and tasks.

3.6
job
overall occupational activity that is carried out by a worker, consisting of all the tasks performed by the worker during the entire working day or shift

Note 1 to entry: A worker often has a job title that describes his or her job, sometimes complemented with an additional description to ensure clear identification, e.g. "welder - process line A".

Note 2 to entry: Figure 1 illustrates the hierarchy of jobs and tasks.



Key

Job 1 scaffolders
Job 2 welders
Job 3 painters
Job 4 store keepers

Task 1 planning
Task 2 grinding
Task 3 welding

Figure 1 — An example illustrating the hierarchy of jobs and tasks

4 Symbols

Symbol	Description	Unit
c_i	sensitivity coefficient related to each input quantity	
c_1	sensitivity coefficient associated with job noise level sampling	
$c_{1a,m}$	sensitivity coefficient associated with noise level sampling of task m	
$c_{1b,m}$	sensitivity coefficient associated with estimation of duration of task m	
c_2	sensitivity coefficient associated with measurement instrumentation	

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Symbol	Description	Unit
c_3	sensitivity coefficient associated with microphone position	
i	task sample number	
I	total number of task samples	
j	number of observations of task duration	
J	total number of observations of task duration	
k	coverage factor related to a confidence interval	
K_N	denominator as given in C.3.3, Note 2	
$L_{\mathrm{EX,8h}}$	A-weighted noise exposure level normalized to a nominal 8 h working day	dB
$\bar{L}_{\mathrm{EX,8h}}$	A-weighted noise exposure level normalized to a nominal 8 h working day averaged over a number of days	dB
$L_{\mathrm{EX,8h,95\,\%}}$	A-weighted noise exposure level including its associated uncertainty - 95 % level of confidence NOTE Statistically, assuming a normal distribution of the noise exposure measurements, for 95 % of the samplings the noise exposure will be equal to or below this level	dB
$L_{\mathrm{EX,8h},m}$	A-weighted noise exposure level of task m contributing to the daily noise exposure level	dB
$L_{p,\mathrm{A,eq}T}$	A-weighted equivalent continuous sound pressure level over a period T	dB
$L_{p,\mathrm{A,eq}T_m}$	A-weighted equivalent continuous sound pressure level for task m	dB
$L_{p,\mathrm{A,eq}T_m,i}$	A-weighted equivalent continuous sound pressure level of sample i for task m	dB
$L_{p,\mathrm{A,eq}T_n}$	A-weighted equivalent continuous sound pressure level of job sample n	dB
$L_{p,\mathrm{A,eq}T_\mathrm{e}}$	A-weighted equivalent continuous sound pressure level for the effective duration of the working day	dB
$L_{p,\mathrm{Cpeak}}$	C-weighted peak sound pressure level	dB
m	task number	
M	total number of tasks	
n	Job/day sample numbers	
N	total number of job/day samples	
n_G	number of workers in a homogeneous noise exposure group	
p_0	reference sound pressure; $p_0 = 2 \times 10^{-5}$ Pa	Pa
p_A	A-weighted sound pressure	Pa
p_Cpeak	C-weighted peak sound pressure	Pa
Q_2	correction for measurement instrumentation	dB
Q_3	correction for microphone position	dB
T	time period over which an average is taken	h
T_0	reference duration; $T_0 = 8$ h	h
T_e	effective duration of the working day	h
T_m	duration of task m	h
$T_{m,j}$	duration of sample j of task m	h
T_n	duration of sample n (job or full-day approaches)	h
U	expanded uncertainty	dB
u	combined standard uncertainty	dB
u_i	standard uncertainty of each input quantity	dB
u_1	standard uncertainty of the energy average of a number of measurements of A-weighted equivalent continuous sound pressure level	dB
$u_{1a,m}$	standard uncertainty due to noise level sampling of task m	dB

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Symbol	Description	Unit
$u_{1b,m}$	standard uncertainty due to the estimation of duration of task m	h
u_2	standard uncertainty due to the instrumentation	dB
$u_{2,m}$	standard uncertainty due to the instrumentation in the task method	dB
u_3	standard uncertainty due to microphone position	dB
$\bar{U}$	expanded uncertainty for multiple nominal days	dB
x	day number for multiple nominal days	
X	total number of days for multiple nominal days	

5 Instrumentation

5.1 Sound level meters and personal sound exposure meters

Measurements can be made by using either integrating-averaging sound level meters or personal sound exposure meters.

Sound level meters, including the microphone and associated cables, shall meet the requirements for IEC 61672-1, class 1 or class 2 instrumentation. Class 1 instrumentation is preferred and should be used when measuring at very low temperatures or when the noise is dominated by high frequencies (see also NOTES 2 and 4).

Personal sound exposure meters, including the microphone and cable, shall meet the requirements specified in IEC 61252. Personal sound exposure meters also fulfilling the requirements of IEC 61672-1, class 1, are recommended when measuring at very low temperatures or when the noise is dominated by high frequencies (see also NOTES 2 and 4).

NOTE 1 Personal sound exposure meter is often referred to as "noise dose meter" or "noise dosimeter".

NOTE 2 For IEC 61672-1, class 1 instruments, the specified tolerance limits are applied for the temperature range from -10 °C to +50 °C. For instrumentation in accordance with IEC 61672-1, class 2, and for personal sound exposure meters in accordance with IEC 61252, the influence of variations in the air temperature on the measured signal level is specified over the range from 0 °C to +40 °C. In order to maintain accuracy when performing measurements outside this temperature range, it can be necessary to use an instrument for which the manufacturer specifies compliance for a wider temperature range. Alternatively, a sound level meter in accordance with IEC 61672-1, class 1, can be selected. In cold conditions, the measuring instrument can be kept warm, e.g. under clothing, such that only the microphone is exposed to low temperatures.

NOTE 3 The choice of the instrumentation influences the uncertainty of the measurements.

NOTE 4 For personal sound exposure meters, IEC 61252 allows wide tolerances in the frequency characteristics above 4 000 Hz, which can lead to incorrect measurement of high frequency sound such as that from air nozzles. In order to reduce the uncertainty when measuring noise dominated by high frequencies, it can be necessary to use a measuring instrument for which the manufacturer specifies high frequency characteristics within a narrower tolerance range. Alternatively, a sound level meter specified in accordance with IEC 61672-1, class 1 or 2, can be selected in the following way. Due to the tolerances defined in IEC 61672-1, class 2 sound level meters can be appropriate for measurements up to 8 kHz. Measurements up to 16 kHz can only be performed by using a sound level meter meeting the requirements of class 1. Assessment of airborne sound beyond 16 kHz can produce unreliable results even when using class 1 sound level meters (see also Clause 1).

Personal sound exposure meters can have a cut-off level at around 70 dB. This can influence the value of the measured noise exposure, but only at low exposure levels.

5.2 Calibrator

For class 1 measuring instrument, the sound calibrator shall conform to class 1 specifications of IEC 60942.

For class 2 measuring instrument, the sound calibrator shall conform to either class 1 or 2 specifications of IEC 60942.

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5.3 Periodic verification

The calibration of the sound calibrator and the compliance of the integrating-averaging sound level meter with the requirements of IEC 61672-1, IEC 61252 and other relevant standards shall be verified at intervals in a laboratory making calibrations traceable to appropriate standards.

Unless national regulations dictate otherwise, it is recommended that the sound calibrator and the compliance of the integrating-averaging sound level meter with the requirements of IEC 61672-1 be verified at intervals not exceeding 2 years.

The date for the last periodic verification, the name of the laboratory that performed it, and the outcome shall be recorded and given in the measurement report.

6 Methodology – Chronological steps

6.1 Step 1: Work analysis

The work analysis shall provide sufficient information about the work and the workers under consideration so that an appropriate measurement strategy can be selected and measurements can be planned. Work analysis shall be carried out in accordance with Clause 7.

6.2 Step 2: Selection of the measurement strategy

A measurement strategy shall be selected from task-based measurement, job-based measurement and full-day measurement as specified in Clause 8. More than one measurement strategy can be used, if relevant (see B.6).

6.3 Step 3: Measurements

The basic measurement quantity shall be $L_{p,\mathrm{A,eq}T}$. In addition, $L_{p,C_{\mathrm{peak}}}$ shall be measured, if relevant. The measurements shall follow the chosen strategy as specified in one of Clauses 9, 10 or 11 and comply with the requirements of Clause 12.

6.4 Step 4: Error handling and uncertainties

Sources of errors and uncertainties that shall influence the result shall be evaluated in accordance with Clauses 13 and 14.

6.5 Step 5: Calculation and presentation of results and uncertainty

Calculate $L_{\mathrm{EX,8h}}$ as specified for the selected strategy (see Clauses 9, 10, and 11) and the uncertainty as specified in Annex C. The results and uncertainties can be calculated by using the spreadsheet in https://standards.iso.org/iso/9612/ed-3/en.

The results shall be presented as specified in Clause 15. Annexes D, E and F provide practical examples for the task-based, job-based, and full-day measurements, respectively.

7 Work analysis

7.1 General

Work analysis is required in all situations. It shall provide the information necessary to

a) describe the activities of the enterprise and the jobs of the workers under consideration,

b) define homogeneous noise exposure groups (see 7.2), if relevant,

c) determine a nominal day or several nominal days, as appropriate, for each worker or group,

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d) identify tasks which make up the jobs, if relevant,

e) identify possible significant noise events,

f) choose the measurement strategy, and

g) establish the measurement plan.

The work shall be analysed with emphasis put on production, process, organization, workers and activities.

The measurements shall be performed by using the task-based, job-based or full-day strategy. Whichever strategy is used, it is essential to identify all events which are significant with regard to noise exposure and to make sure that the measurement plan takes them into account. See Annex A for an example of a checklist.

NOTE The order in which the items above are performed can depend on the complexity of the situation on site. The items are strongly connected, and the process can therefore be iterative in complex situations, i.e. increased knowledge about one of the items can result in a new description or redefinition of others.

7.2 Defining homogeneous noise exposure groups (HEG)

Measurement efforts can be reduced by defining homogeneous noise exposure groups. These are groups of workers that are performing the same job and are expected to have similar noise exposures during the working day. If used, the homogeneous noise exposure group shall be clearly identified.

NOTE The term "similar noise exposure group" is sometimes used instead of "homogeneous noise exposure group".

Homogeneous noise exposure groups can be defined in a number of ways. For example, it may be possible to define such groups according to job title, function, work area or profession. Alternatively, the groups can be defined by analysing the work according to production process or work activity criteria.

In whichever way the groups are defined, they should be verified in consultation with the workers and supervisors, and ultimately by evaluating the measurement results, see 10.4.

7.3 Determination of a nominal day

A nominal day, including work periods and breaks, shall be determined in consultation with both workers and management. The work shall be studied in order to obtain an overview and understanding of all factors, which can influence the noise exposure. See Annex A for more details.

Issues that shall be addressed are:

a) tasks (content and duration) and variation within tasks;

b) main noise sources and noisy work areas;

c) work pattern and any significant noise events, resulting in a change of the noise level or which can result in peak levels exceeding regional regulation limits;

d) number and duration of breaks, meetings, etc., and whether they should be regarded as a part of the nominal day.

Measurements shall be planned to ensure that all significant noise events are included. For each of the events, it shall be recorded when it occurred, its nature, duration and daily frequency. An example of a checklist to ensure that significant noise events are detected during the work analysis is given in Annex A.

In some cases, the work and consequently the noise exposure, varies from day to day so that there is no typical daily exposure, e.g. for workers who work in different locations or jobs each day. In these cases, the nominal day can be defined from work situations during several days, e.g. one week. See also Notes to 3.2 and 3.3.

Any indicators that characterize the work with respect to noise shall be identified, quantified, and reported. Examples of such indicators are: type of production in process; materials; quantities; thickness of work pieces; adjustment; speed; and number of workers involved.

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If the purpose of measurements is to estimate the long-term risk of hearing impairment of workers, then the nominal day chosen shall be representative of the average exposure over the period under consideration, in accordance with ISO 1999.

8 Selection of measurement strategies

8.1 General

The selection of an appropriate measurement strategy is influenced by several factors such as the purpose of the measurements, complexity of the work situation, number of workers involved, effective duration of the working day, time available for measurement and analysis, and amount of detailed information required.

8.2 Measurement strategies

Three measurement strategies for the determination of workplace noise exposure are offered by this document. These are:

a) task-based measurement: the work performed during the day is analysed and split up into a number of representative tasks, and for each task separate measurements of sound pressure level are taken (see Clause 9);

b) job-based measurement: a number of random samples of sound pressure level are taken during the performance of particular jobs (see Clause 10);

c) full-day measurement: sound pressure level is measured continuously over complete working days (see Clause 11).

Detailed guidance on the choice of the measurement strategy is given in Annex B.

A guidance for the evaluation of the daily noise exposure level of workers who perform several types of work in a same day or over several days is provided in Annex G.

9 Strategy 1 – Task-based measurement

9.1 Dividing the nominal day into tasks

For the workers or homogeneous noise exposure groups under evaluation, the nominal day shall be divided into tasks. Each task shall be defined such that $L_{p,\mathrm{A,eq}T}$ is likely to be repeatable. Care shall be taken to ensure that all relevant noise contributions are included. Detailed information regarding the duration of tasks is particularly important for noise sources with high noise levels.

Identification of the noise sources and tasks that give the highest peak levels is important to obtain a correct determination of both $L_{p,\mathrm{A,eq}T}$ and $L_{p,\mathrm{Cpeak}}$.

The use of a handheld tool can be regarded as a separate task. This will better guide the noise reduction plan and will likely reduce the measurement uncertainty.

9.2 Duration of tasks

The durations of the tasks, T_m, shall be determined. This can be done by:

a) interviewing the workers and the supervisor;

b) observing and measuring durations during noise measurements;

c) gathering information regarding operation of typical noise sources (e.g. work processes, machines, activities at the workplace and in its surroundings).

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Optionally, the duration of a task can be regarded as a variable. To determine possible variations in duration, the task can be observed and the duration recorded, for instance, three times. Alternatively, multiple workers and supervisors can be asked to indicate the most reasonable duration range.

If J observations of the task duration, $T_{m,j}$, are available, the arithmetic average value of task duration, T_m, is given by Formula (1):

$$T_m = \frac{1}{J}\sum_{j=1}^{J} T_{m,j} \tag{1}$$

The sum of individual durations of tasks, T_m, which make up the nominal day, shall correspond to the effective duration of the working day. The effective duration of the working day, T_e, is given by Formula (2):

$$T_e = \sum_{m=1}^{M} T_m \tag{2}$$

where

T_m is the arithmetic average duration of task m;

m is the number of a task;

M is the total number of tasks.

NOTE Task-based measurements can, for instance, be combined with full-day measurements to verify that all relevant sources are included.

9.3 Measurement of $L_{p,\mathrm{A},\mathrm{eq}T_m}$ for tasks

9.3.1 General

For each task, the $L_{p,\mathrm{A},\mathrm{eq}T_m}$ representative of the noise exposure of the worker shall be measured in accordance with Clause 12. The measurements shall cover variations in noise level within each task in time, space and working conditions.

The measurement technician shall ensure that the work situation is representative. The worker under consideration shall be observed during the measurements whenever possible. If the operating or work conditions deviate from the normal situation, this shall be recorded and reported.

In cases when it is difficult for the measurement technician to follow the worker's activities without interfering with them, the activities during the measurements shall be registered by other means, e.g. by interviews and/or by reviewing work record sheets filled in by the worker, and reported.

9.3.2 Number of measurements

To cover as much of the actual variation in noise levels as possible, it is recommended to measure on different workers within a HEG. In this case, at least 5 measurements shall be made for each task. Otherwise, at least 3 measurements shall be made for each task, if just one worker is sampled. A recommendation of the number of workers on which the tasks measurements can be performed in function of the HEG size is given below:

Table 1 — Minimum recommended number of workers, $n_{\min}$, on which measurements can be taken as a function of the size of the HEG, n_G

n_G	1 to 2	3 to 5	6 to 11	12 to 15	16 to 20	Above 21
$n_{\min}$	1	2	3	4	5	6

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If the range for $L_{p,\mathrm{A,eq}T_m,i}$ is larger than 5 dB (if a group of workers is sampled), or larger than 3 dB (if only one worker is sampled), then select one of the following two options:

a) subdivide the task into two tasks, and carry out for both tasks new measurements with the above indicated minimum numbers of samples. Proceed iteratively until the appropriate max-min range limit is complied with for each sub-task.

b) repeat the measurements with a longer duration of each measurement. If the appropriate max-min range limit is then complied with, proceed by discarding the previous set of measurements and retain only the latter set. If the appropriate max-min range is not complied with, then reconsider option a).

9.3.3 Time and duration of measurements

To cover as much of the actual variation in noise levels as possible, it is recommended to measure at different times during the task. Measurement times shall be selected after a careful examination of the work situation, so that the full set of measurements (see 9.3.2) shall be representative of the noise level of the whole task. Splitting a single long measurement into many short measurements, in order to reduce the calculated uncertainty, is therefore not acceptable. This would not ensure representativity of the task noise level, and it would result in an unrealistically low sampling uncertainty (see C.2.3).

As a general rule, the duration of each measurement shall be at least 5 min if the duration of the task is longer than 5 min, or equal to the duration of the task if the latter is shorter than 5 min. The duration of each measurement can, however, be reduced if the sound pressure level is found to be stable, or if the noise from the task is considered to be a minor contributor to the total noise exposure. In order to warrant an accurate average value of $L_{p,\mathrm{A,eq}T_m}$ the minimum and maximum duration of the consecutive measurements of one specific task should not differ by more than 25 %.

NOTE The sound pressure level is considered stable when the overall equivalent sound pressure level does not change by more than 0,2 dB when increasing the measurement duration by 30 s.

If the task noise is cyclic, the duration of each measurement shall correspond to the duration of an integer number (≥3) of full cycles, and not be shorter than 3 min.

9.3.4 Calculation of the A-weighted equivalent continuous sound pressure level

Calculate the A-weighted equivalent continuous sound pressure level for task *m* from *I* separate measurements, $L_{p,\mathrm{A,eq}T_m,i}$, by Formula (3) as follows:

$$L_{p,\mathrm{A,eq}T_m} = 10\lg\left(\frac{1}{I}\sum_{i=1}^{I} 10^{0,1\cdot L_{p,\mathrm{A,eq}T_m,i}}\right)\mathrm{dB} \tag{3}$$

where

$L_{p,\mathrm{A,eq}T_m,i}$ is the ith determination of the A-weighted equivalent continuous sound pressure level for task *m* of duration T_m;

i is the measurement number;

I is the total number of measurements carried out for task *m*.

Calculate the uncertainty in accordance with C.2.

9.4 Calculation of contribution from each task to daily noise exposure level

The calculation specified in this subclause shall be performed if a value for the relative contribution of each task to the daily noise exposure level is required. Otherwise, proceed to 9.5.

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The noise contribution from task *m* to the daily A-weighted noise exposure level, $L_{\mathrm{EX,8h,m}}$, can be calculated from Formula (4):

$$L_{\mathrm{EX,8h,m}} = L_{p,\mathrm{A,eq}T_m} + 10\lg\left(\frac{T_m}{T_0}\right)\mathrm{dB} \tag{4}$$

where

$L_{p,\mathrm{A,eq}T_m}$ is the A-weighted equivalent continuous sound pressure level for task *m* as given by Formula (3);

T_m is the arithmetic average duration in hours of task m as given by Formula (1);

T_0 is the reference duration in hours of the working day, T_0 = 8 h.

This intermediate result can help to identify which tasks contribute most to $L_{\mathrm{EX,8h}}$ and support decisions with regard to effective reduction of exposure to noise.

9.5 Determination of daily noise exposure level

Calculate the daily A-weighted noise exposure level $L_{\mathrm{EX,8h}}$ from Formula (5) or from Formula (6).

Formula (5) allows the calculation of daily A-weighted noise exposure level from the $L_{p,\mathrm{A,eq}T_m}$ and the duration of each of the tasks. It uses the level calculated from Formula (3) and the duration in accordance with 9.2.

$$L_{\mathrm{EX,8h}} = 10\lg\left(\sum_{m=1}^{M}\frac{T_m}{T_0}10^{0,1\,L_{p,\mathrm{A,eq}T_m}}\right)\mathrm{dB} \tag{5}$$

where

$L_{p,\mathrm{A,eq}T_m}$ is the A-weighted equivalent continuous sound pressure level for task *m* as given by Formula (3);

T_m is the arithmetic average duration in hours of task *m* as given by Formula (1);

T_0 is the reference duration in hours of the working day, T_0 = 8 h;

m is the task number;

M is the total number of tasks *m* contributing to the daily noise exposure level.

Formula (6) allows the calculation of the A-weighted noise exposure level from the noise contribution of each of the tasks. It shall be used if the relative contribution from each task m has been calculated in accordance with 9.4 using Formula (6):

$$L_{\mathrm{EX,8h}} = 10\lg\left(\sum_{m=1}^{M}10^{0,1\,L_{\mathrm{EX,8h},m}}\right)\mathrm{dB} \tag{6}$$

where

$L_{\mathrm{EX,8h},m}$ is the A-weighted noise exposure level of task m contributing to the daily noise exposure level;

m is the task number;

M is the total number of tasks *m* contributing to the daily noise exposure level.

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10 Strategy 2 – Job-based measurement

10.1 General

The principle of this measurement strategy is that random samples of noise exposure are taken by measuring $L_{p,\mathrm{A,eq}T}$ during the performance of jobs identified during the work analysis.

10.2 Measurement plan – Number, duration and distribution of measurements

The measurement plan shall be established as follows. From the jobs identified, homogeneous noise exposure groups shall be established (see also 7.2). For each homogeneous noise exposure group:

a) Determine from Table 2 the minimum number of workers, $n_{\min}$, to be sampled according to the size n_G of the HEG:

Table 2 — Minimum number of workers, $n_{\min}$, to be sampled

n_G	1 to 2	3 to 5	6 to 9	10 to 14	15 to 20	21 to 30	31 to 50	51 to 75	76 to 100	>100
$n_{\min}$	1	2	3	4	5	7	9	12	15	20

b) Determine from Table 3 the minimum cumulative duration of measurements over the number of workers, n_G, of the homogeneous exposure group:

Table 3 — Minimum cumulative duration of the measurements to be distributed over the HEG

Number of workers in the HEG	Minimum cumulative duration of the measurements
$n_G \leq 15$	6 h + (n_G -1) × 1 h
$15 < n_G \leq 40$	20 h + (n_G -15) × 45 min
$n_G > 40$	40 h or split the group

c) Select a sample duration. The recommended sample duration is 45 min. The duration shall be at least 15 min and not greater than 1 h.

Samples shall be of the same duration and have to be distributed randomly among the members of the group and across the duration of the working day.

EXAMPLE A measurement plan was to be made for a homogeneous exposure group of six workers. The measurement plan was as follows:

— three workers (in accordance with Table 2) were randomly chosen among the six workers;

— the minimum cumulative measurement duration of samples was determined to be 11 h (in accordance with Table 3);

— the number of samples was chosen to be 15, which resulted in a duration of 45 min each;

— the first sample was chosen to start right at the beginning of the working day and the last one to include the end of the working day because work analysis had shown that these periods have a significant contribution to noise exposure;

— the thirteen other samples were spread randomly over the rest of the working day.

10.3 Measurements

Measurements shall be carried out in accordance with Clause 12.

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10.4 Determination of daily noise exposure levels for workers in a homogeneous noise exposure group

Calculate the A-weighted equivalent continuous sound pressure level, $L_{p,\mathrm{A,eq}T_\mathrm{e}}$, for the effective duration of the working day, T_e, from Formula (7):

$$L_{p,\mathrm{A,eq}T_\mathrm{e}} = 10\lg\left(\frac{1}{N}\sum_{n=1}^{N} 10^{0,1\cdot L_{p,\mathrm{A,eq}T_n}}\right)\mathrm{dB} \tag{7}$$

where

$L_{p,\mathrm{A,eq}T_n}$ is the A-weighted equivalent continuous sound pressure level of the nth sample;

n is the sample number;

N is the total number of samples.

Calculate the daily A-weighted noise exposure level, $L_{\mathrm{EX,8h}}$, of the workers in a given homogeneous noise exposure group from Formula (8):

$$L_{\mathrm{EX,8h}} = L_{p,\mathrm{A,eq}T_\mathrm{e}} + 10\lg\left(\frac{T_\mathrm{e}}{T_0}\right)\mathrm{dB} \tag{8}$$

where

$L_{p,\mathrm{A,eq}T_\mathrm{e}}$ is the A-weighted equivalent continuous sound pressure level for the effective duration of the working day;

T_e is the effective duration of the working day;

T_0 is the reference duration, T_0 = 8 h.

Calculate the uncertainty in accordance with C.3.

If the uncertainty contribution c_1u_1 due to sampling (as obtained from Table C.4) is greater than 3,5 dB, then the uncertainty shall be decreased by:

a) Modifying the homogeneous exposure group to lower the standard uncertainty u_1. Measurements carried out before this modification takes place can still be used if they correspond to the newly assembled HEG.

b) Performing additional measurements to increase the number of samples.

11 Strategy 3 — Full-day measurement

11.1 General

Full-day measurement shall cover all work-related noise contributions and quiet periods during the working day (see NOTE 1). It is practical to perform these long-duration measurements by using personal sound exposure meters.

When using this measurement strategy, it shall be ensured that the chosen days are representative for what is defined as the relevant work situation (see NOTE 2).

For practical reasons, if it is not possible to measure the entire working day, then measurements should be made over as large a part of the day as possible, covering all significant periods of noise exposure.

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Measurements shall cover at least 75 % of the nominal daily working time and shall be representative of the noise exposure over the entire work shift.

NOTE 1 Noisiest events will strongly influence the result. It is also important not to neglect the quiet periods as this avoids overestimating the noise exposure. For example, an 8 h day can start with a 1 h briefing and end with a 1 h tidying up and dressing. In this case, the working days contains 2 h without relevant exposure. If the other 6 h are spent in noise (85 dB with grinding, milling, cutting, drilling, etc.), considering only these 6 h leads to a daily exposure of 85 dB versus 83,8 dB if the quiet phases are included.

NOTE 2 Since this measurement strategy collects all contributions, it also has the highest risk of including false contributions (see Clause 13). This risk can be reduced by carefully observing the worker during measurements, by taking spot measurements and/or by asking the worker at the end of the shift about the tasks he/she performed or the locations he/she worked at.

11.2 Observing work activities and monitoring measurements

Workers should be observed during the measurements. If this is not possible, the validity of the measurements should be checked by one or more of the following actions:

a) interviewing supervisors and workers;

b) taking spot measurements to verify the levels measured using personal sound exposure meters;

c) assessing the exposure of selected workers using the task-based measurements as specified in Clause 9;

d) an examination, by the worker and the measurement technician, of the personal sound exposure meter log (time history) at the end of the shift, in order to identify the different tasks and events. For this reason, the use of logging personal sound exposure meters is highly recommended.

11.3 Measurements

The number of persons to be sampled in the HEG shall be determined according to Table 4.

Table 4 — Minimum number of workers n_{min} to be sampled in a HEG of n_G size

n_G	1 to 2	3 to 5	6 to 9	10 to 14	15 to 20	21 to 30	31 to 50	51 to 75	76 to 100	> 100
n_{min}	1	2	3	4	5	7	9	12	15	20

Measurements shall be carried out in accordance with Clause 12. A minimum of three full-day measurements and at least one measurement on every worker to be sampled in the HEG is required. If the results of the three measurements differ by less than 3 dB and c_1u_1 does not exceed 3,5 dB. c_1u_1 shall be determined using Table C.4. Calculate the A-weighted equivalent continuous sound pressure level during the nominal day as the energy average of the three measurements. For calculation, see Formula (8).

If the results of the three measurements differ by 3 dB or more, or if c_1u_1 exceeds 3,5 dB, take at least two additional full-day measurements, and calculate the A-weighted equivalent continuous sound pressure level during the nominal day as the energy average of all measurements.

In all cases the uncertainty contribution c_1u_1 due to sampling shall not exceed 3,5 dB. If this is not met even after two additional full-day measurements, then modifications to the homogeneous noise exposure group should be made or the number of measurements should be increased to meet this requirement on c_1u_1. Note that the number of samples is equal to the number of days.

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11.4 Determination of daily noise exposure level

Calculate the daily A-weighted noise exposure level, $L_{\mathrm{EX,8h}}$, from Formula (9):

$$L_{\mathrm{EX,8h}} = L_{p,\mathrm{A,eq}T_{\mathrm{e}}} + 10\lg\left(\frac{T_{\mathrm{e}}}{T_0}\right)\mathrm{dB} \tag{9}$$

where

$L_{p,\mathrm{A,eq}T_{\mathrm{e}}}$ is the A-weighted equivalent continuous sound pressure level for the effective duration of the working day;

T_{e} is the effective duration of the working day;

T_0 is the reference duration, T_0 = 8 h.

Calculate the uncertainty in accordance with C.4.

12 Measurements

12.1 Selection of instrumentation

Measurements shall be performed using the following types of instrumentation (see also 5.1):

a) personal sound exposure meter worn by the worker whose noise exposure is being determined;

b) integrating-averaging sound level meter used according to 12.4.

Personal sound exposure meters can be used for measurements in all types of work situations. It is the preferred method when making long duration measurements for a mobile worker engaged in complex or unpredictable tasks or carrying out a large number of discrete tasks.

For measurements of single or multiple tasks at fixed workstations, hand-held or fixed sound level meters can be used.

12.2 Field calibration

Field calibration includes an acoustic calibration check of the entire measuring system, including the microphone, and is a distinct survey procedure separate from laboratory calibration. A field calibration shall consist of applying a sound calibrator meeting the requirements of 5.2 to each microphone and recording the measured level at one or more frequencies within the frequency range of interest. It is recommended to perform this calibration in an ambient sound pressure level not exceeding the ambient sound level recommended by the manufacturer of the calibrator.

Before each series of measurements and at the start of each daily series of measurements, a field calibration with appropriate adjustment shall be performed. At the end of each series of measurements and at the end of each daily series of measurements, a field calibration without adjustment shall be performed. If the reading at any frequency at the end of a series of measurements differs from the reading of that frequency at the beginning of the series by more than 0,5 dB, the results of the series of measurements shall be discarded.

12.3 Instrument worn by the worker

Instrument worn by the worker can be personal sound exposure meter or an integrating sound level meter using microphone extension cable.

The measuring instrument shall be started according to the manufacturer's instructions. This shall be done after the calibration has been performed, the measuring instrument has been attached and the microphone has been installed in place to ensure that no extraneous noises are introduced while fitting the instrument. The starting time of the measurement shall be noted. When the measurements are finished,

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the instrumentation shall be stopped according to the manufacturer's instructions before the removal of instrumentation and microphone. The time when the measurement was stopped shall be noted.

The microphone shall be mounted on the top of the shoulder at a distance of at least 0,1 m from the entrance of the external ear canal at the side of the most exposed ear and should be approximately 0,04 m above the shoulder. The microphone and the cable, if any, shall be fastened in such a way that mechanical influence or covering by clothing do not lead to false results.

Care shall be taken not to disturb the work performance and especially not to introduce safety risks. Similarly, care shall be taken to avoid false contributions. See also 13.2.

NOTE When using measuring instruments worn by the worker or in other cases when the microphone is placed very close to a worker's body, the measurement result is affected by screening effects and reflections by the body. This is especially valid for high-frequency noise and small noise sources at a short distance from the ear. In such cases, the measurements can be performed with microphones placed at both sides of the head in order to establish the exposure of the most exposed ear.

The worker being monitored shall be informed of the purpose of the measurement. Workers shall be advised not to remove the measuring instrument during the entire measurement period and to perform their work normally.

Any high peak sound levels recorded by the instrument which were not validated by observation shall be investigated and commented in the report.

12.4 Integrating-averaging sound level meter

The measured levels shall be representative of the noise level at the worker's ear. If the sound field is uniform, precise measurement position is less critical.

Measurements should be made with the microphone positioned at the locations of the worker's head during normal performance of the job or task. Preferably, the microphone shall be placed at the centre plane of the worker's head, on a line with the eyes, with its axis parallel to the worker's vision, and without the worker present. Various relevant space positions of the worker's head shall be taken into account. The average sound pressure level at the workstation can also be found by moving the sound level meter around (by sweeping) in the area of interest. The sweep can be performed by moving the microphone at constant speed along a path shaped like the infinity sign: ∞ .

In cases where the worker has to be present at his/her workstation, the microphone shall be placed or held at a distance between 0,1 m and 0,4 m from the entrance of the external ear canal and at the side of the most exposed ear.

If the worker's activity or the workstation configuration makes it impossible to keep the distance within 0,4 m, the use of an instrument worn by the worker is recommended.

If the worker's location is very close to the noise sources, the sound field should be carefully investigated and the microphone position and direction chosen shall be precisely stated in the test report.

If the head position at a workstation is not well defined, the following microphone heights shall be used (see ISO 11200[2], ISO 11201[3], ISO 11202[4], ISO 11203[5] and ISO 11205[6]):

a) standing worker: 1,55 m ± 0,075 m above the ground on which the worker is standing;

b) seated worker: 0,8 m ± 0,05 m above the middle of the seat plane with the seat set at or as near as possible to the midpoint of its horizontal and vertical adjustment.

Even if the worker is working at a fixed workstation, measurements with a fixed microphone position can give an over or underestimation of the real exposure in cases where the worker moves about the machine. In these cases, a personal sound exposure meter should be used.

Close to the sound source, even small changes of the microphone position can result in significant variations of the sound pressure level. If tones are clearly audible at the workstation, standing sound waves can occur. To determine the local variation of the sound pressure level, the microphone should be moved through the

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relevant area of occupation close to the source. The variations of the sound pressure level observed during the movement of the microphone are treated as time-varying levels and averaged accordingly. If a scan is used for investigating the variations in an area, the noise level as a function of time, $L(t)$, should be measured with a suitable time resolution. Supplementary measurements with an instrument fixed to the worker in accordance with 12.3 reduce the uncertainty associated with the noise exposure from this source.

Special measurement procedures are required for the measurement of noise exposure underneath earphones (e.g. for secretaries, telephonists, pilots, air traffic controllers) or underneath helmets (e.g. pilot and motorcycle helmets). For noise sources close to the ear, measurements in the ear canal should be performed in accordance with ISO 11904-1[7] or ISO 11904-2[8].

13 Sources of uncertainty and errors

13.1 General

Some sources of uncertainty or error need specific consideration in order to reduce their influence as far as possible.

The main sources of uncertainty in the result are:

a) variations in the daily work, operating conditions, uncertainty in sampling, etc.;

b) instrumentation and calibration;

c) microphone position;

The most typical sources of error are:

d) false contributions, for instance from wind, airflows or impact on the microphone and the microphone rubbing on clothing;

e) lacking or faulty work analysis or poor definition of nominal day;

f) contributions from non-typical noise sources and non-typical behaviour.

Item a) depends on the complexity of the work situation. These variations are expected to be the highest for a mobile worker among non-constant noise sources. Item b) depends on where the microphone is fixed and what class of instrumentation and calibrator is used. Items c), d), and e) should be reduced by good practice as specified in this document. Regarding item f), such possible sources of uncertainty shall be identified during work analysis and it shall be decided whether they are to be included or not.

The relevant noise contributions shall be identified during the work analysis and the measurements. If significant contribution from sources of error is detected, the measurements shall be rejected or corrected.

The measured noise exposure level and the associated uncertainty in the result depend on the measurement method used. A measurement made by an instrument worn by the worker can include contributions from non-representative sources which can increase the measured noise exposure level. The use of a hand-held sound level meter ensures that such sources are excluded, e.g. by pausing the measurement. Measurements at fixed positions can lead to an underestimation of the worker's noise exposure due to the difficulty in handling noise contributions from sound sources close to the ear, such as hand-held tools.

13.2 Mechanical impacts on microphone

Errors due to mechanical impacts on the microphone shall be avoided by ensuring that the microphone or the windscreen is not being touched or hit by anything. This effect can be more difficult to observe when using a personal sound exposure meter. Many measuring instruments record high sound pressure levels if the microphone and/or the connecting devices are touched, rubbed or otherwise come into contact with other objects.

The measurements shall be checked for such uncertainty by comparing the detailed measurement result (log) where available, with observations during the measurements (e.g. notes in a handwritten log). If

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unexplained peak levels occur, the influence on the result shall be investigated, and the measurement shall be excluded if the result is significantly affected. In cases where non-logging instruments are used and unexplained high peak levels occur, the measurement shall be repeated.

13.3 Wind and airflows

Measurements in airflows with high flow velocity, e.g. wind, shall be avoided. If this is not possible, care shall be taken to minimize the flow-induced noise. If possible, the contribution of flow noise should be investigated by measurements made in similar work situations without airflows. If not, the airflow-induced noise can be assessed by measuring in places without the occupational noise but with similar flow.

Airflow-induced noise shall be reduced by using a windscreen on the microphone. For personal sound exposure meters, the size of the windscreen is usually limited. By using a hand-held sound level meter with a larger windscreen, the potential effect of airflow-induced noise can be controlled.

Hand-held sound level meters should be equipped with windscreens of a minimum 60 mm diameter to reduce the effect of airflow or wind.

NOTE Contributions from wind and airflows depend on the wind speed and the size of the windscreen. A-weighted sound pressure levels around 80 dB are usually not significantly influenced by airflow speeds up to 10 m/s, provided the windscreen is of 60 mm diameter or more.

13.4 Relevance of sound contributions

Care shall be taken in defining the sounds that are relevant for the noise exposure. Noise from sources such as radios, speech, and alarm signals shall be considered as relevant if work analysis has shown that they are part of the normal work condition. However, if the person who carries out the measurement has good reasons for considering such a contribution as being irrelevant, he/she shall exclude it from the measured data, provided that this is reported.

If non-typical behaviour is observed at the workplace during measurements, an evaluation of possible influences on the measurement result shall be done. If the influence is regarded as significant, new measurements shall be performed.

14 Calculation of measurement uncertainties and presentation of the final results

The uncertainties associated with measuring occupational noise exposure shall be determined in accordance with Annex C and Annex H if appropriate.

The expanded measurement uncertainty, together with the corresponding coverage factor, shall be stated for a level of confidence of 95 %.

The $L_{\mathrm{EX,8h}}$ and the value of its associated uncertainty U shall be given as well as the final result $L_{\mathrm{EX,8h,95\,\%}}$ as defined in Formula (10):

$$L_{\mathrm{EX,8h,95\,\%}} = L_{\mathrm{EX,8h}} + U \text{ dB} \tag{10}$$

15 Information to be reported

The report of noise exposure measurements carried out in accordance with this document shall provide the following information:

a) general information:

   1) name of the client (company, department, etc.) of the investigation;

   2) identification of the working place(s) (building, workshop, etc.);

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3) identification of the worker(s) or group(s) of workers (such as name or worker number) whose exposure has been determined;

4) name of the person(s) and company or institution who carried out measurements and calculations;

5) purpose of the determination;

6) reference to this document and the strategy that has been applied;

b) work analysis:

1) description of the work activities investigated;

2) size and composition of homogeneous noise exposure groups, where relevant;

3) description of the day(s) investigated, including the tasks comprising the nominal day when the task-based measurements have been made;

4) measurement strategy/strategies employed, together with a reference to the statistical approach used;

c) instrumentation:

1) identification and class of instrumentation used (manufacturer, model, serial number);

2) configuration of the system, e.g. windscreen, extension cable, etc.;

3) calibration traceability (date and result of the most recent verification of the components of the measuring system);

4) documentation of calibration checks performed before and after each measurement;

d) measurements:

1) identification of worker(s) whose noise exposure was measured;

2) date and time of measurements;

3) instrumentation used for each measurement (if various instruments are used);

4) description of work undertaken by the worker during the course of the measurements, including duration of work activity and, if relevant, duration of cyclic events contained within the work activity;

5) report of any deviations from the normal work conditions or normal work behaviour during the course of the measurements;

6) production indicators related to the work being undertaken, where relevant;

7) description of the sources of noise contributing to the noise exposure;

8) description of any irrelevant sounds included in or deleted from the measured results;

9) description of any events observed which can have influenced the measurements (e.g. airflows, impacts on the microphone, impulsive noise);

10) relevant information on meteorological conditions (e.g. wind, rain, temperature);

11) position and orientation of microphone(s);

12) number of measurements at each position;

13) duration of each measurement;

14) duration of each task in the nominal day, and the associated uncertainty, when using the task-based approach;

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15) results of each measurement, to include at least the $L_{p,\mathrm{A,eq}T}$ and, optionally, the highest $L_{p,\mathrm{Cpeak}}$ values;

e) results and conclusions:

1) A-weighted equivalent continuous sound pressure level $L_{p,\mathrm{A,eq}T}$ and, optionally, C-weighted peak sound pressure level $L_{p,\mathrm{Cpeak}}$ for each task/job;

2) the number of peak events exceeding the regional regulation for $L_{p,\mathrm{Cpeak}}$, and the description of these specific noise events;

3) when using the task-based measurement, the values of $L_{\mathrm{EX,8h},m}$ for each task, if relevant;

4) A-weighted noise exposure level $L_{\mathrm{EX,8h}}$ for the nominal day(s), and the highest C-weighted peak sound pressure level $L_{p,\mathrm{Cpeak}}$ if measured during all tasks, rounded to one decimal place;

5) expanded uncertainty U (or $\bar{U}$ as appropriate) associated with $L_{\mathrm{EX,8h}}$ (or $\bar{L}_{\mathrm{EX,8h}}$ as appropriate) and $L_{p,\mathrm{Cpeak}}$, if available, for the nominal day(s), rounded to one decimal place (noise exposure and the measurement uncertainty shall be reported as separate values);

6) $L_{\mathrm{EX,8h,95\,\%}}$ which is the sum of the A-weighted noise exposure level $L_{\mathrm{EX,8h}}$ or $\bar{L}_{\mathrm{EX,8h}}$ (as appropriate) and its associated expanded uncertainty U or $\bar{U}$ (as appropriate);

7) where necessary, it should be added the information on the decision rule applied and statement on conformity of the results with requirements such as national action or limit values.

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Annex A
(informative)

Example of a checklist to ensure that significant noise events are detected during the work analysis

	Yes	No
Are any of these situations encountered?		
— use of compressed air jets	☐	☐
— compressed air releases	☐	☐
— hammering	☐	☐
— intense impacts	☐	☐
— occasional use of very noisy machines and tools	☐	☐
— noisy vehicles passing by	☐	☐
Are there very noisy operations during particular phases?		
— at the beginning of shift	☐	☐
— at the end of shift	☐	☐
— during adjustment, supply phases	☐	☐
— during start-up and shut-down activities or production	☐	☐
— during cleaning phases	☐	☐
— other	☐	☐

Are there very noisy activities at neighbouring workstations?

— type: _____

— exposed workstations: _____

See also 7.3.

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Annex B
(informative)

Guidance to the selection of measurement strategy

B.1 General

This annex provides details of basic measurement strategies and guidance on selecting the one most appropriate (see Clause 8).

B.2 Strategy 1 — Task-based measurement

This strategy focuses on the tasks producing significant noise exposure and on minimizing the measurement duration required for a specified uncertainty. Task-based measurement is most useful when the work can be split into well-defined tasks with clearly definable noise conditions during which measurements can be made. Care should, however, be taken to ensure that all relevant noise contributions are included in the measurement period, which also requires knowledge of any short-duration high-exposure acoustical events during the working day.

The strategy is based on a detailed work analysis in order to understand all tasks. Further, it requires a continuous validation of the measurements. This allows a smaller number of measurements to be made of the tasks producing little variation in noise level.

Task-based measurements provide information on the contributions from the different tasks to the daily noise exposure. This is advantageous if the purpose of the measurements is to determine priorities for a noise control program. It also opens the possibility of calculating the noise exposure for working days different from the measurement day as regards the distribution and durations of tasks. It can reduce the measurement effort compared to the other methods.

The use of this strategy gives significant time savings of measurement when large groups of workers are doing similar activities in similar acoustical environments. The measurements can also be more easily controlled. Similarly, when the organization or the content of the work has changed, a new assessment of noise exposure will only require partial re-examination when the task method is used.

If the work situation is complex, the work analysis can be time-consuming.

B.3 Strategy 2 — Job-based measurement

Job-based measurements are most useful when typical work patterns and tasks are hard to describe or when it is not desirable or practical to perform a detailed work analysis. It is not recommended to use this method if a job consists of a small number of very noisy tasks.

Job-based measurements can result in a reduced effort required for work analysis. Care should be taken in defining the jobs to ensure that the noise exposure of any worker within a given job is representative. This strategy can be time-consuming due to the time required for the measurements, but produces a smaller uncertainty in the result obtained compared to the full day measurement strategy.

As with task-based measurement, care should be taken to ensure that major noise contributions are included in the measuring period. Job-based measurements do not necessarily provide any information on the relative contribution of different tasks within a job to the daily noise exposure, since they do not take into account tasks being carried out within the defined job.

In most cases, this strategy requires a longer measurement duration than the task-based strategy.

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B.4 Differences and similarities between task-based and job-based measurement

B.4.1 General: These two strategies are not mutually exclusive. Task-based and job-based measurements are both based on sampling of noise levels. A given work situation can, in many cases, be handled using one strategy or the other with the same quality of result. The main differences between these strategies are described in B.4.2 to B.4.4.

B.4.2 Different homogeneous noise exposure groups: For job-based measurements, the homogeneous noise exposure groups can have a different content than for task-based measurements. As a job is a group of tasks performed by one worker, job-based measurement does not require a detailed breaking down of the work activity into tasks. Establishing homogeneous noise exposure groups is therefore less time-consuming than in the case of task-based measurements. Task-based measurements also require a better knowledge of the work situation than the other strategies, so that all tasks that contribute to the noise level are clarified along with their respective durations.

B.4.3 Different measurement plans: For job-based measurements, the measurement plan is normally easier to implement than for task-based measurements because there is no need to isolate each task that has to be measured during the working day.

B.4.4 Different durations of measurements: Job-based measurement requires longer measurement periods than task-based measurement.

B.5 Strategy 3 — Full-day measurement

Full-day measurement is, like job-based measurement, most useful when typical work patterns and tasks are hard to describe. However, it requires even less effort in analysing the work. On the other hand, if the work situation is simple, this strategy can require a longer measurement duration than any of the others.

Full-day measurement is recommended when the noise exposure pattern of workers is unknown, unpredictable or complex. It is also possible to use this strategy for any pattern of noise exposure, especially where it is not necessary or desirable to perform a detailed work analysis.

Full-day measurement can also be useful to verify that all the major contributions are included. For such checking purposes, the daily noise exposure level can be determined directly without any additional calculations.

By using a logging instrument, information about the fluctuations in noise level during the working day can be obtained, and it shall be possible to determine the contributions from different tasks. It also gives the possibility to remove irrelevant sound contributions from the measurement results. Use of a logging instrument is therefore highly recommended when performing full-day measurements.

Measurements of long duration are most likely to be performed with a personal sound exposure meter or similar instrumentation that is worn by the worker whose noise exposure is being measured. In these cases, there is a high likelihood that measurements contain some artefacts that are not related to the typical noise exposure of the worker, e.g. impacts on the microphone (accidental or deliberate), or deliberate interference, such as shouting by work colleagues or attempts to work in a deliberately noisy manner. For these reasons, it is highly recommended that the measurements be directly observed by the measurement technician, or that some other equally effective means of identifying and accounting for these artefacts be used. Measurements that are unobserved are highly prone to such measurement artefacts, in which case the best approach shall be to perform measurements over a number of days in order for the “novelty” of the measurement process to diminish.

B.6 Using more than one measurement strategy

There are occasions when it is necessary or desirable to use more than one measurement strategy. For instance, if the nominal day tends to be complex, the calculated noise exposure from task-based

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measurements can be checked by performing full-day measurements of selected workers. If the result differs by more than the uncertainty a further study of the nature of the exposure is required.

There can be situations during full-day or task-based measurements where some tasks are not performed, although they are a part of the nominal day. In those cases, additional measurements of those tasks are necessary.

As a further example, some workers work in different patterns during a day. This can require the use of task-based measurement for work in the morning, and job-based measurement for work in the afternoon. The measurement made with the job-based strategy is then considered as a long duration task.

B.7 Selection of measurement strategy for different work patterns

Table B.1 provides guidance to the selection of the basic measurement strategy depending on the work pattern.

Table B.1 — Selection of basic measurement strategy

Type or pattern of work	Measurement strategy		
	Strategy 1 Task-based measurement	Strategy 2 Job-based measurement	Strategy 3 Full-day measurement
Fixed workstation — Simple or single task	✓[a]	—	—
Fixed workstation — Complex or multiple tasks	✓[a]	✓	✓
Mobile worker — Predictable pattern — Small number of tasks	✓[a]	✓	✓
Mobile worker — Predictable work — Large number of tasks or complex work pattern	✓	✓	✓[a]
Mobile worker — Unpredictable work pattern	—	✓	✓[a]
Fixed or mobile worker — Multiple tasks with unspecified duration of tasks	—	✓[a]	✓
Fixed or mobile worker — No tasks assigned	—	✓[a]	✓

✓ Strategy can be used.

[a] Recommended strategy.

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Annex C
(normative)

Evaluation of measurement uncertainties

C.1 General

This annex specifies the procedure for determining the expanded uncertainty of the A-weighted noise exposure level normalized to an 8 h working day, $L_{\mathrm{EX,8h}}$ or, alternatively, of the measured value of the A-weighted equivalent continuous sound pressure level, $L_{p,\mathrm{A,eq}T}$.

The procedure is in accordance with ISO/IEC Guide 98-3. The uncertainties can be calculated by using the spreadsheet in https://standards.iso.org/iso/9612/ed-3/en.

Strict fulfilment of the requirements of this document, in particular those relating to the avoidance of faulty contributions to the calculated noise exposure level (see Clause 13), ensures that no systematic errors due to such faulty contributions are present in the final result.

Sources of uncertainty considered in this annex are indicated in Table C.1.

The uncertainty due to the selection of measurement day(s) is not included in Table C.1, as in many cases this can only be determined by extensive measurements over longer periods. However, strict fulfilment of the requirements of this document, in particular regarding work analysis, keeps this uncertainty under control.

Table C.1 — Sources of uncertainty considered in determining the expanded uncertainty of A-weighted equivalent continuous sound pressure levels or noise exposure levels normalized to an 8 h working day

Source of uncertainty	Application	Subscript[a]	Clause
Sampling of task noise levels	Task-based measurement	1a	C.2
Estimation of task durations	Task-based measurement	1b	C.2
Sampling of job noise levels	Job-based measurement	1	C.3
Instrumentation	All strategies	2	C.5
Microphone location	All strategies	3	C.6

NOTE Table C.1 covers uncertainties listed against items a), b), and c) in 13.1. When the measurements are performed in accordance with this document, it is assumed that the uncertainty due to errors listed in 13.1 d), e), and f) is reduced to be insignificant and/or included in the noise level sampling.

[a] Used in symbols for partial uncertainties and sensitivity coefficients.

In cases where there is evidence that a source of uncertainty not considered in this annex is playing a major role, its contribution can be included in the calculation of the combined standard uncertainty by adding lines to Table C.2 or Table C.3.

The contributions to the combined standard uncertainty, u, associated with the value of the noise exposure level depends on the standard uncertainty, u_i, of each input quantity and the related sensitivity coefficients, c_i. The sensitivity coefficients are measures of how the noise exposure level is affected by changes in the values of the respective input quantities. Mathematically, they are equal to the partial derivatives of the functional relationship [see Formulae (C.2) or (C.8) and ISO/IEC Guide 98-3] with respect to the relevant input quantity. The contributions of the respective input quantities are given by the products of the standard

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uncertainties and their associated sensitivity coefficients. The combined standard uncertainty, u, is obtained from the individual uncertainty contributions, $c_i u_i$, using Formula (C.1):

$$u^2 = \sum c_i^2 u_i^2 \tag{C.1}$$

The expanded uncertainty, U, is given by $U = k \cdot u$, where k is a coverage factor that is a function of the confidence interval. For the purpose of this document, one-sided confidence interval of 95 % is considered, which results in k = 1,65. This means that 95 % of the values are below the upper limit, $\left[L_{\mathrm{EX,8h}} + U\right]$.

This document allows the use of statistical approaches for calculating the uncertainty other than the methods specified in this annex, e.g. methods based on scientific assessment (expert knowledge about uncertainties) or Monte Carlo simulations. If such methods are used, they shall be in accordance with ISO/IEC Guide 98-3. It shall also be demonstrated that they do not underestimate the uncertainty. The method used shall be indicated in the test report.

NOTE 2 An explanation of the statistical background of the methods given in this annex is given in References [11] and [12].

C.2 Determination of the expanded uncertainty for task-based measurement

C.2.1 Functional relationship for task-based measurement

The general expression for the determination of the A-weighted noise exposure level, $L_{\mathrm{EX,8h}}$, using task-based measurement is given by Formula (C.2):

$$L_{\mathrm{EX,8h}} = 10 \lg \left[\sum_{m=1}^{M} \frac{T_m}{T_0} 10^{0,1\, L^*_{p,\mathrm{A,eq}T_m}} \right] \mathrm{dB} \tag{C.2}$$

where

T_m is the arithmetic average duration of task m;

T_0 is the reference duration, T_0 = 8 h;

m is the task number;

M is the total number of tasks;

$L^*_{p,\mathrm{A,eq}T_m}$ is the estimate of the true A-weighted equivalent continuous sound pressure level for task m, $L_{p,\mathrm{A,eq}T_m}$ [see Formula (3)]:

$$L^*_{p,\mathrm{A,eq}T_m} = L_{p,\mathrm{A,eq}T_m} + Q_2 + Q_3$$

in which

Q_2 is the correction for the measuring instrumentation used for the determination of the A-weighted continuous sound pressure level,

Q_3 is the correction for the microphone position used for the determination of the A-weighted equivalent continuous sound pressure level.

NOTE As the estimates of both Q_2 and Q_3 are approximately 0, $L^*_{p,\mathrm{A,eq}T_m} \approx L_{p,\mathrm{A,eq}T_m}$. Under these conditions, Formula (C.2) is identical to Formula (5).

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C.2.2 Calculation of the combined standard uncertainty, u, and the expanded uncertainty, U

Considering that quantities involved are not correlated, the combined standard uncertainty for the A-weighted noise exposure level $L_{\mathrm{EX,8h}}$, $u(L_{\mathrm{EX,8h}})$ shall, in accordance with ISO/IEC Guide 98-3, be calculated from the numerical values of uncertainty contributions, $c_i u_i$, as follows by Formula (C.3):

$$u^2(L_{\mathrm{EX,8h}}) = \sum_{m=1}^{M} \left[c_{1a,m}^2 \left(u_{1a,m}^2 + u_{2,m}^2 + u_3^2 \right) + c_{1b,m}^2 u_{1b,m}^2 \right] \quad \text{(C.3)}$$

where

$u_{1a,m}$	is the standard uncertainty due to noise level sampling of task m, see C.2.3;
$u_{1b,m}$	is the standard uncertainty due to the estimation of duration of task m, see C.2.3;
$u_{2,m}$	is the standard uncertainty due to the instrumentation used for task m;
u_3	is the standard uncertainty due to microphone position;
$c_{1a,m}$ and $c_{1b,m}$	are the corresponding sensitivity coefficients for task m;
m	is the task number;
M	is the total number of tasks.

The expanded uncertainty is $U = 1{,}65 \times u$.

NOTE 1 Due to the linear relation between the measured noise level and the estimate for the noise level, the sensitivity coefficients for instrumentation, $c_{2,m}$, microphone location, $c_{3,m}$, and noise level sampling, $c_{1a,m}$, have the same values, i.e. $c_{2,m} = c_{3,m} = c_{1a,m}$. The coefficients $c_{2,m}$ and $c_{3,m}$ are therefore replaced by $c_{1a,m}$ in Formula (C.3).

NOTE 2 Formula (C.3) is strictly valid for the case where M A-weighted equivalent continuous sound pressure levels are determined using different instrumentation for each measurement. Nevertheless, since the main uncertainty contributions of instrumentation, such as influences of level linearity, frequency response of the microphone, angle of sound incidence and spectral weighting, are different in different sound field positions for the same instrumentation, it is assumed that Formula (C.3) is appropriate.

The corresponding uncertainty budget is shown in Table C.2.

Table C.2 — Uncertainty budget for determining noise exposure levels for task-based measurement

Quantity	Estimate	Standard uncertainty u_i	Probability distribution	Sensitivity coefficient c_i	Uncertainty contribution $c_i\, u_i$ dB
$L_{p,\mathrm{A,eq}T_m}$	Energy average of measured $L_{p,\mathrm{A,eq}T_m}$ for task m	$u_{1a,m}$ for each task, to be determined using Formula (C.6)	Normal	$c_{1a,m}$ for each task, to be determined using Formula (C.4)	$c_{1a,m} \cdot u_{1a,m}$ 1 value per task
T_m	Estimated value of duration T_m for task m	$u_{1b,m}$ for each task, to be determined using Formula (C.7)	Normal	$c_{1b,m}$ for each task, to be determined using Formula (C.5)	$c_{1b,m} \cdot u_{1b,m}$ 1 value per task
Q_2	0	$u_{2,m}$ as given by Table C.5	Normal	$c_{2,m} = c_{1a,m}$	$c_{1a,m} \cdot u_{2,m}$
Q_3[a]	0	u_3 as given in C.6	Normal	$c_{3,m} = c_{1a,m}$	$c_{1a,m} \cdot u_3$

[a] It is expected that Q_3 will be in the range from −1,0 dB to 0,5 dB. For simplicity, the arithmetic average estimate of Q_3 is taken equal to zero. The standard uncertainty related to microphone positions, u_3, is assumed to cover this extra uncertainty.

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C.2.3 Contributions to measurement uncertainty and the uncertainty budget

For task-based measurement, the sensitivity coefficients are as follows by Formula (C.4) and Formula (C.5):

$$c_{1a,m} = \frac{\partial L_{\mathrm{EX,8h}}}{\partial L^{*}_{p,\mathrm{A,eq}T_m}} = \frac{T_m}{T_0} 10^{0,1\left(L^{*}_{p,\mathrm{A,eq}T_m} - L_{\mathrm{EX,8h}}\right)} \tag{C.4}$$

$$c_{1b,m} = \frac{\partial L_{\mathrm{EX,8h}}}{\partial T_m} = 4,34 \frac{c_{1a,m}}{T_m} \tag{C.5}$$

The standard uncertainty, $u_{1a,m}$, in noise level due to sampling for task m is given by Formula (C.6):

$$u_{1a,m} = \sqrt{\frac{1}{I(I-1)}\left[\sum_{i=1}^{I}(L_{p,\mathrm{A,eq}T_m,i} - L_{p,\mathrm{A,eq}T_m})^2\right]} \tag{C.6}$$

where

$L_{p,\mathrm{A,eq}T_m}$ is the arithmetic average of I measured A-weighted equivalent continuous sound pressure levels for task m, i.e. $L_{p,\mathrm{A,eq}T_m} = \frac{1}{I}\sum_{i=1}^{I} L_{p,\mathrm{A,eq}T_m,i}$;

i is the number of task sample;

I is the total number of task samples.

The standard uncertainty, $u_{1b,m}$, due to duration of task m can be calculated from the measured durations from independent measurements as follows by Formula (C.7):

$$u_{1b,m} = \sqrt{\frac{1}{J(J-1)}\left[\sum_{j=1}^{J}(T_{m,j} - T_m)^2\right]} \tag{C.7}$$

where J is the total number of observations of task duration.

NOTE If a time range has come out of the work analysis, an estimate is $u_{1b,m} = 0,5(T_{\max} - T_{\min})$.

C.3 Determination of the expanded uncertainty for job-based measurement

C.3.1 Functional relationship for job-based measurement

The general expression for determination of the daily A-weighted noise exposure level, $L_{\mathrm{EX,8h}}$, using job-based measurement is as given by Formula (C.8):

$$L_{\mathrm{EX,8h}} = 10 \lg \frac{T_{\mathrm{e}}}{T_0}\left[\sum_{n=1}^{N} 10^{0,1\, L^{*}_{p,\mathrm{A,eq}T_n}}\right] \mathrm{dB} \tag{C.8}$$

where

T_{e} is the effective duration of the working day;

T_0 is the reference duration, T_0 = 8 h;

n is the number of job sample;

N is the total number of job samples;

$L^{*}_{p,\mathrm{A,eq}T_n}$ is the estimate of the true A-weighted equivalent continuous sound pressure level for the job sample n, $L_{p,\mathrm{A,eq}T_n}$:

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$$L^{*}_{p,\mathrm{A},\mathrm{eq}T_n} = L_{p,\mathrm{A},\mathrm{eq}T_n} + Q_2 + Q_3$$

in which

Q_2 is the correction for the measurement instrumentation used for determination of the A-weighted equivalent continuous sound pressure level,

Q_3 is the correction for the microphone position used for determination of the A-weighted equivalent continuous sound pressure level.

NOTE As the estimates of both Q_2 and Q_3 are approximately 0, $L^{*}_{p,\mathrm{A},\mathrm{eq}T_n} \approx L_{p,\mathrm{A},\mathrm{eq}T_n}$. Under these conditions, Formula (C.8) gives the same result as Formulae (7) and (8).

C.3.2 Calculation of the combined standard uncertainty, u, and the expanded uncertainty, U

The combined standard uncertainty for the A-weighted noise exposure level $L_{\mathrm{EX},8\mathrm{h}}$, $u(L_{\mathrm{EX},8\mathrm{h}})$, shall, in accordance with ISO/IEC Guide 98-3, be calculated from the numerical values of all uncertainty contributions, $c_j u_j$, taken from Table C.3 as follows Formula (C.9):

$$u^2\left(L_{\mathrm{EX},8\mathrm{h}}\right) = c_1^2 u_1^2 + c_2^2 u_2^2 + c_3^2 u_3^2 \tag{C.9}$$

The expanded uncertainty is $U = 1{,}65 \times u$.

NOTE Formula (C.9) is strictly valid for the case where N A-weighted equivalent continuous sound pressure levels are determined using different instrumentation for each measurement. Nevertheless, since the main uncertainty contributions of instrumentation, such as the influences of level linearity, the frequency response of the microphone, the angle of sound incidence and the spectral weighting, are different in different sound field positions for the same instrumentation, it is assumed that Formula (C.9) is appropriate.

C.3.3 Contributions to measurement uncertainty and the uncertainty budget

For job-based measurement:

— The uncertainty contribution, $c_1 u_1$, of job noise level sampling is given in Table C.4 as a function of the number, N, of job noise level samples and the standard uncertainty, u_1, of measured values $L_{p,\mathrm{A},\mathrm{eq}T,n}$.

— The sensitivity coefficients, c_2 and c_3, for the uncertainty due to the instrumentation and that due to imperfect selection of measurement position, respectively, are as given by Formula (C.10) and Formula (C.11):

$$c_2 = 1 \tag{C.10}$$

$$c_3 = 1 \tag{C.11}$$

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Table C.3 — Uncertainty budget for determination of noise exposure levels for job-based measurement

Quantity	Estimate	Standard uncertainty u_i	Probability distribution	Sensitivity coefficient c_i	Uncertainty contribution $c_i u_i$ dB
$L_{p,\mathrm{A,eq}T}$	$L_{p,\mathrm{A,eq}T}$ energy average of the measured $L_{p,\mathrm{A,eq}T_n}$	u_1 to be determined using Formula (C.12)	Normal	c_1	$c_1 u_1$ as given by Table C.4
Q_2	0	u_2 as given by Table C.5	Normal	$c_2 = 1$	u_2
Q_3[a]	0	u_3 as given in C.6	Normal	$c_3 = 1$	u_3

[a] It is expected that Q_3 will be in the range from −1,0 dB to 0,5 dB. For simplicity, the arithmetic average estimate of Q_3 is taken equal to zero. The standard uncertainty related to microphone positions, u_3, is assumed to cover this extra uncertainty.

The uncertainty contribution, $c_1 u_1$, of job noise level sampling is obtained directly from the energy average of the measured values of the job noise level samples, $L_{p,\mathrm{A,eq}T_n}$, and the standard uncertainty, u_1, of these values using Table C.4.

The standard uncertainty, u_1, is given by Formula (C.12):

$$u_1 = \sqrt{\frac{1}{(N-1)}\left[\sum_{n=1}^{N}\left(L_{p,\mathrm{A,eq}T_n} - \bar{L}_{p,\mathrm{A,eq}T}\right)^2\right]} \qquad \text{(C.12)}$$

where

$L_{p,\mathrm{A,eq}T_n}$ is the A-weighted equivalent continuous sound pressure level for job noise level sample n;

$\bar{L}_{p,\mathrm{A,eq}T}$ is the arithmetic average of N job samples of the A-weighted continuous equivalent sound pressure level, i.e. $\bar{L}_{p,\mathrm{A,eq}T} = \frac{1}{N}\sum_{n=1}^{N} L_{p,\mathrm{A,eq}T_n}$;

n is the number of job sample;

N is the total number of job samples;

NOTE Formula (C.12) is used to calculate u_1, the entry value of Table C.4. The resulting u_1 is denoted here as standard uncertainty to maintain similar terminology for all u_1 terms, but usually it is denoted standard deviation.

Table C.4 — Uncertainty contribution, $c_1 u_1$, of job and full-day noise level sampling, in decibels, applicable to a set of N measured values, $L_{p,\mathrm{A,eq}T_n}$, of standard uncertainty u_1,

u_1	Uncertainty contribution $c_1 u_1$ of measured values $L_{p,\mathrm{A,eq}T_n}$ dB											
N	0,5	1	1,5	2	2,5	3	3,5	4	4,5	5	5,5	6
3	0,6	1,6	3,1	**5,2**	**8,0**	**11,5**	**15,7**	**20,6**	**26,1**	**32,2**	**39,0**	**46,5**
4	0,4	0,9	1,6	2,5	**3,6**	**5,0**	**6,7**	**8,6**	**10,9**	**13,4**	**16,1**	**19,2**
5	0,3	0,7	1,2	1,7	2,4	3,3	**4,4**	**5,6**	**6,9**	**8,5**	**10,2**	**12,1**
6	0,3	0,6	0,9	1,4	1,9	2,6	3,3	**4,2**	**5,2**	**6,3**	**7,6**	**8,9**
7	0,2	0,5	0,8	1,2	1,6	2,2	2,8	3,5	**4,3**	**5,1**	**6,1**	**7,2**
8	0,2	0,5	0,7	1,1	1,4	1,9	2,4	3,0	**3,6**	**4,4**	**5,2**	**6,1**
9	0,2	0,4	0,7	1,0	1,3	1,7	2,1	2,6	3,2	**3,9**	**4,6**	**5,4**

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Table C.4 *(continued)*

u_1	Uncertainty contribution c_1u_1 of measured values $L_{p,\mathrm{A,eq}T_n}$ dB											
N	0,5	1	1,5	2	2,5	3	3,5	4	4,5	5	5,5	6
10	0,2	0,4	0,6	0,9	1,2	1,5	1,9	2,4	2,9	3,5	**4,1**	**4,8**
12	0,2	0,3	0,5	0,8	1,0	1,3	1,7	2,0	2,5	2,9	3,5	**4,0**
14	0,1	0,3	0,5	0,7	0,9	1,2	1,5	1,8	2,2	2,6	3,0	3,5
16	0,1	0,3	0,5	0,6	0,8	1,1	1,3	1,6	2,0	2,3	2,7	3,2
18	0,1	0,3	0,4	0,6	0,8	1,0	1,2	1,5	1,8	2,1	2,5	2,9
20	0,1	0,3	0,4	0,5	0,7	0,9	1,1	1,4	1,7	2,0	2,3	2,6
25	0,1	0,2	0,3	0,5	0,6	0,8	1,0	1,2	1,4	1,7	2,0	2,3
30	0,1	0,2	0,3	0,4	0,6	0,7	0,9	1,1	1,3	1,5	1,7	2,0

When c_1u_1 as obtained from Table C.4 is greater than 3,5 dB (values indicated in **bold** in Table C.4), it is recommended that the measurement plan be revised or changed to reduce u_1 (see 10.4).

NOTE 1 The values for N = 3 and N = 4 are given only for use with full-day measurements (see Clause C.4).

NOTE 2 In field situations where there is a need to evaluate whether more measurements are necessary, a simpler estimation of u_1 can be made. The estimated standard uncertainty, u_1^* can be calculated by using the following formula:

$$u_1^* = \frac{L_{p,\mathrm{A,eq}T_n}(\max) - L_{p,\mathrm{A,eq}T_n}(\min)}{K_N}$$

where

K_N = 2,2 if $N < 6$

K_N = 2,5 if $6 \leq N \leq 15$

K_N = 3,0 if $16 \leq N \leq 30$

NOTE 3 For intermediate values of u_1, the uncertainty contribution c_1u_1 can be obtained using the spreadsheet provided with the standard.

NOTE 4 Example for the use of the table C.4: if a measurement includes eight samples (N=8) and the standard deviation u_1 in those eight results is 4,5 dB, then the uncertainty contribution due to sampling c_1u_1 is 3,6 dB and the maximum value of 3,5 is exceeded.

C.4 Uncertainty calculation for the full-day measurement

The procedure for calculation of uncertainty for full-day measurement is the same as for the job method. Thus, the uncertainty for full-day measurement is found from the uncertainty budget in Table C.3 and by using Formula (C.9) with c_1u_1 from Table C.4, and u_2 and u_3 from Clauses C.5 and C.6.

C.5 Standard uncertainty, u_2, for the instrumentation used

The standard uncertainty, u_2 (or $u_{2,\mathrm{m}}$ for the task m), of instrumentation is given in Table C.5.

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Table C.5 — Standard uncertainty, u_2, of instrumentation

Type of instrumentation	Standard uncertainty u_2(or $u_{2,\mathrm{m}}$) dB
Sound level meter as specified in IEC 61672-1, class 1	0,7
Personal sound exposure meter as specified in IEC 61252	1,5
Sound level meter as specified in IEC 61672-1, class 2	1,5

NOTE 1 The standard uncertainty listed in Table C.5 is only valid for $L_{p,\mathrm{A,eq}T}$.

NOTE 2 The standard uncertainties, u_2 (or $u_{2,\mathrm{m}}$), given in Table C.5 are based on empirical data. Experience shows that these standard uncertainty values of instrumentation are representative for most relevant situations.

NOTE 3 If properly justified and reported the investigator can use a different value than that reported in Table C.5 for u_2.

C.6 Standard uncertainty, u_3, due to measurement position

The standard uncertainty, u_3, due to measurement position is 1,0 dB.

NOTE The given value for standard uncertainty is based on empirical data. For cases when the microphone is worn by the worker and when the microphone is placed close to the worker's body, the uncertainty is due to screening effects and reflections from the body. For cases when measurements are performed with the worker absent, the uncertainty is due to the microphone position(s) not being fully representative of the true position(s) of the worker's head. See also 12.3 and 12.4 with regard to microphone position.

C.7 Uncertainty for multiple nominal days

In case of multiple nominal days are defined according to 7.3 then the daily exposure is calculated with Formula (C.13) and the associated uncertainty is calculated according to Formula (C.14).

$$\bar{L}_{\mathrm{EX,8h}} = 10\lg\frac{1}{N}\sum_{x=1}^{X} n_x\, 10^{0,1\,L_{\mathrm{EX,8h},x}}\ \mathrm{dB}\ \text{ where }\ N = \sum_{x=1}^{X} n_x \qquad \text{(C.13)}$$

where

X is the number of nominal days considered;

n_x is the number of days among N days during which the worker is working according to nominal day x;

$L_{\mathrm{EX,8h},x}$ is the noise exposure for nominal day x.

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$$\bar{U} = \sqrt{\sum_{x=1}^{X} \left(\frac{n_x \,.\, 10^{0,1\, L_{\mathrm{EX,8h},x}}}{\sum_{y=1}^{X} n_y \,.\, 10^{0,1\, L_{\mathrm{EX,8h},y}}} \right)^2 U_x^{\,2}} \qquad \text{(C.14)}$$

where

n_y is the number of days among N days during which the worker is working according to nominal day y;

U_x is the expanded uncertainty associated to $L_{\mathrm{EX,8h},x}$;

$L_{\mathrm{EX,8h},y}$ is the noise exposure for nominal day y.

NOTE Regional regulation shall limit the maximum number of days for the calculation of the daily exposure. As a general guidance it is recommended not to average for more than a week.

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Annex D
(informative)

Example showing calculation of daily noise exposure level using task-based measurements

D.1 Step 1: Work analysis

In this example, the noise exposure level for welders working in a mechanical workshop is determined by using task-based measurements. The working day consists of the following sequence of tasks:

— work planning (quiet);

— two periods of cutting, grinding and welding of steel plates;

— lunch (in this case, the lunch is regarded as part of the working day);

— work planning (quiet);

— two periods of cutting, grinding and welding of steel plates.

All welders perform the same work and therefore can be regarded as one homogeneous noise exposure group.

According to information from the supervisor, the work can be split into three separate tasks: welding, cutting and grinding. The other tasks (breaks and planning) do not contribute to noise exposure.

The welders say they spend between 1 h and 2 h a day cutting and grinding, and between 4 h and 6 h on a day welding. The rest of the day is dedicated to planning and breaks.

Based on this information, the nominal day is defined in Table D.1.

Table D.1 — Welder's nominal day

Task	Duration h
Work planning, breaks (quiet)	1,5
Cutting and grinding	1,5
Welding	5,0
Total	8,0
NOTE The time spent on each task is calculated using the average of the range of values given by the welder and his or her supervisor.	

D.2 Step 2: Selection of strategy

Since the number of tasks is limited and well-defined, the situation is suitable for a task-based measurements.

D.3 Step 3: Measurements

The contribution to noise exposure from work planning and breaks is negligible. A few simple noise measurements with a sound level meter are sufficient to establish that fact. In this example, the equivalent noise level in these periods is considered to be $L_{p,\mathrm{A,eq}T}$ = 70 dB, which is a conservative estimate (sound level meter observations give values less than or equal to this threshold).

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Since the noise contribution from both grinding and welding is highly dependent on the location of the worker's ears in relation to the noisy tool, it was decided to use a personal sound exposure meter for the measurement of these tasks.

It is recommended (9.3.3, 9.3.2) that the measurement period cover at least three work cycles, if the task is cyclic, and be representative of the tasks. Observations have shown that the measurement duration for noise from grinding should be at least 7 min. Similarly, the measurement duration for noise from welding is set to 5 min.

Only one worker is measured. Therefore, three measurements are required for each task (9.3.2). However, since the noise from quiet activities is negligible, only some brief samples of noise level are made during these tasks.

The first measurements resulted in the following values:

Noise level during planning and breaks	$L_{p,\mathrm{A,eq}T_{1-1}}$ < 70 dB		
Noise levels from welding	$L_{p,\mathrm{A,eq}T_{2-1}}$ = 80,1 dB	$L_{p,\mathrm{A,eq}T_{2-2}}$ = 82,2 dB	$L_{p,\mathrm{A,eq}T_{2-3}}$ = 79,6 dB
Noise levels from cutting and grinding	$L_{p,\mathrm{A,eq}T_{3-1}}$ = 86,5 dB	$L_{p,\mathrm{A,eq}T_{3-2}}$ = 92,4 dB	$L_{p,\mathrm{A,eq}T_{3-3}}$ = 89,3 dB

Since the difference between the measured noise levels from cutting and grinding exceeds 3 dB, those three first measurements are discarded. Then the measurement duration was increased to 15 min and the number of samples extended to five and distributed on two workers, giving the following results:

New measurements of noise levels from cutting and grinding (dB)	$L_{p,\mathrm{A,eq}T_{3-1}}$ 92	$L_{p,\mathrm{A,eq}T_{3-2}}$ 87,2	$L_{p,\mathrm{A,eq}T_{3-3}}$ 92,1	$L_{p,\mathrm{A,eq}T_{3-4}}$ 88,3	$L_{p,\mathrm{A,eq}T_{3-5}}$ 88,7

D.4 Step 4: Error handling

In this example, observations during the measurements showed that there were no significant risks of making measurement errors.

D.5 Step 5: Calculation and presentation of the results including uncertainty

D.5.1 Calculation of daily A-weighted noise exposure level

The noise level from each task is calculated using Formula (3) including all measurement results. The noise level from welding is then as follows:

$$L_{p,\mathrm{A,eq}T_2} = 10\lg\left[\frac{1}{3}\left(10^{80,1/10} + 10^{82,2/10} + 10^{79,6/10)}\right)\right]\mathrm{dB} = 80,8\ \mathrm{dB}$$

Similarly, the noise level from cutting and grinding was found to be 90,1 dB.

The noise level during planning and breaks was initially defined to be 70 dB.

The contribution to the daily A-weighted noise exposure level is calculated for each activity according to Formula (4). The contributions from the tasks to the daily noise exposure are then:

a) planning and breaks

$$L_{\mathrm{EX,8h,1}} = 70\ \mathrm{dB} + 10\lg\left(\frac{1,5}{8}\right)\mathrm{dB} = 62,7\ \mathrm{dB}$$

b) welding

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$$L_{\mathrm{EX,8h,2}} = 80{,}8\ \mathrm{dB} + 10\lg\left(\frac{5}{8}\right)\mathrm{dB} = 78{,}8\ \mathrm{dB}$$

c) cutting and grinding

$$L_{\mathrm{EX,8h,3}} = 90{,}1\ \mathrm{dB} + 10\lg\left(\frac{1{,}5}{8}\right)\mathrm{dB} = 82{,}8\ \mathrm{dB}$$

The daily A-weighted noise exposure level can now be calculated from Formula (6):

$$L_{\mathrm{EX,8h}} = 10\lg\left(10^{62{,}7/10} + 10^{78{,}8/10} + 10^{82{,}8/10}\right)\mathrm{dB} = 84{,}3\ \mathrm{dB}$$

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D.5.2 Calculation of uncertainty

The standard uncertainty, $u_{1a,2}$, due to sampling of welding noise levels is calculated from Formula (C.6).

$$u_{1a,2} = \sqrt{\frac{1}{2\times 3}\left[(-0{,}5)^2 + (1{,}6)^2 + (-1{,}0)^2\right]} = 0{,}8\ \text{dB}$$

Similarly, the standard uncertainty due to the noise level sampling from cutting and grinding is found to be $u_{1a,3}$ = 1,01 dB. The standard uncertainty due to planning and breaks can be set to 0 dB since these tasks do not contribute to the combined standard uncertainty, u.

A personal sound exposure meter as specified in IEC 61252 has been used. Therefore, according to Table C.5, the standard uncertainty due to instrumentation is $u_{2,m}$ = 1,5 dB.

In accordance with C.6, the standard uncertainty due to the microphone position is u_3 = 1,0 dB.

The sensitivity coefficients for uncertainty due to noise level sampling, instrumentation, and measurement position are calculated from Formula (C.4).

For planning and breaks, with $L_{p,\mathrm{A,eq}T,1-1}$ < 70 dB, the sensitivity coefficient is:

$$c_{1a,1} = \frac{1{,}5}{8}\,10^{\frac{(70{,}0-84{,}3)}{10}} = 0{,}007 \approx 0$$

For welding, the sensitivity coefficient is:

$$c_{1a,2} = \frac{5}{8}\cdot 10^{\frac{(80{,}8-84{,}3)}{10}} = 0{,}28$$

For cutting and grinding, the sensitivity coefficient is:

$$c_{1a,3} = \frac{1{,}5}{8}\cdot 10^{\frac{(90{,}1-84{,}3)}{10}} = 0{,}71$$

When the uncertainty in duration is excluded, the combined standard uncertainty is found from Formula (C.3), omitting the last part in the brackets.

$$u^2\left(L_{\mathrm{EX,8h}}\right) = 0{,}28^2\cdot\left(0{,}8^2 + 1{,}5^2 + 1{,}0^2\right) + 0{,}71^2\cdot\left(1{,}01^2 + 1{,}5^2 + 1{,}0^2\right) = 2{,}46\ \text{dB}$$

In accordance with C.2.2, the expanded uncertainty, $U\left(L_{\mathrm{EX,8h}}\right)$, is

$$U\left(L_{\mathrm{EX,8h}}\right) = 1{,}65\cdot u = 1{,}65\cdot\sqrt{2{,}46} = 2{,}6\ \text{dB}$$

When the uncertainty in duration is included, the standard uncertainty, $u_{1b,2}$, due to duration of task is given by Formula (C.7). For welding, the standard uncertainty, in hours, due to duration is:

$$u_{1b,2} = \sqrt{\frac{1}{2}\left(1^2 + 1^2\right)} = 1{,}0$$

The sensitivity coefficient related to the uncertainty due to duration is calculated by using Formula (C.5).

The uncertainty due to the duration of welding is $u_{1b,2}$ = 1h.

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The sensitivity coefficient, in decibels times reciprocal hours, for welding is:

$$c_{1b,2} = 4{,}34 \cdot \frac{0{,}28}{5} = 0{,}24$$

The uncertainty due to the duration of cutting and grinding is $u_{1b,3}$ = 0,5 h.

The sensitivity coefficient, in decibels times reciprocal hours, for cutting and grinding is:

$$c_{1b,3} = 2{,}06$$

The combined standard uncertainty, $u\left(L_{\mathrm{EX,8h}}\right)$, can now be calculated using Formula (C.3):

$$u^2\left(L_{\mathrm{EX,8h}}\right) = 0{,}28^2 \cdot \left(0{,}8^2 + 1{,}5^2 + 1{,}0^2\right) + 0{,}71^2 \cdot \left(1{,}01^2 + 1{,}5^2 + 1{,}0^2\right) + (0{,}24 \cdot 1{,}0)^2 + (2{,}06 \cdot 0{,}5)^2 = 3{,}57$$

In accordance with C.2.2, the expanded uncertainty, $U\left(L_{\mathrm{EX,8h}}\right)$, is:

$$U\left(L_{\mathrm{EX,8h}}\right) = 1{,}65 \cdot u = 1{,}65 \cdot \sqrt{3{,}57} = 3{,}1\ \mathrm{dB}$$

D.6 Conclusions

The welders are subject to a daily A-weighted noise exposure level of 84,3 dB, with an associated expanded uncertainty for a one-sided coverage probability of 95 % (k = 1,65) of 2,5 dB, if the uncertainty in duration is omitted or 3,1 dB if this uncertainty is included.

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Annex E
(informative)

Example showing calculation of daily noise exposure level using job-based measurements

E.1 General

In this example, the noise exposure level for production line workers is determined from job-based measurements. Several automated production lines with no major technical differences operate in the plant.

E.2 Step 1: Work analysis

Production line workers perform the same work: running and controlling a production line and intervening in the case of a production incident. Their work includes many tasks (e.g. material supply, production control, product removal, adjustments). However, no distinctions have been found possible between the tasks during the work analysis for the following reasons: the noise exposure conditions of the workers are similar from task to task and the daily duration of each task cannot be determined from work descriptions. The workers make up a homogeneous noise exposure group numbering 12. The effective duration of the working day for this homogeneous noise exposure group is 7,5 h.

E.3 Step 2: Selection of strategy

From the work analysis for this homogeneous noise exposure group of 12 workers, it is neither practical nor desirable to perform a detailed task analysis. Therefore, job-based measurements are selected.

E.4 Step 3: Measurements

The choice of a measurement plan is guided by the following specifications:

— the total minimum duration of the measurements is given in Table 1: for a group of size 12, it amounts to 17 h;

— noise level samples measurements must be distributed over at least 4 workers.

From this, it is decided to take 18 measurements and to set the measurement duration to 1 h.

The distribution of the 18 measurements among the workers in this homogeneous noise exposure group and over the working duration is made knowing that:

a) two personal sound exposure meters are available;

b) working periods for the group are 05:00 to 13:00 and 13:00 to 21:00.

Six workers are selected at random from the 12 members of the homogeneous noise exposure group. The selected distribution of the measurements is as follows:

Day 1: Morning team, 2 workers; measurement periods: 07:00 to 13:00 (six randomly drawn samples);

Day 2: Morning team, 2 different workers; measurement periods: 07:00 to 12:00 (six randomly drawn samples);

Day 2: Afternoon team, 2 different workers; measurement periods: 14:00 to 18:00 (six randomly drawn samples).

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The 18 measurements result in the following values of $L_{p,\mathrm{A,eq}T_n}$:

88,1 dB	86,1 dB	89,7 dB	86,5 dB	91,1 dB	86,7 dB
84,3 dB	78,9 dB	90,1 dB	84,5 dB	87,8 dB	80,2 dB
83,3 dB	87,1 dB	83,2 dB	79,2 dB	78,2 dB	85,6 dB

E.5 Step 4: Error handling

No potential sources of errors are found.

E.6 Step 5: Calculation and presentation of the results and uncertainty

E.6.1 Calculation of daily A-weighted noise exposure level and uncertainty

The energy average of measured values $L_{p,\mathrm{A,eq}T_\mathrm{e}}$ is calculated from Formula (7) as follows:

$$L_{p,\mathrm{A,eq}T_\mathrm{e}} = 10\lg\left(\frac{1}{N}\sum_{n=1}^{N} 10^{0,1\cdot L_{p,\mathrm{A,eq}T_n}}\right) = 86,5\,\mathrm{dB}$$

The standard uncertainty of measured values is:

u_1 = 3,9 dB

See Formula (C.12).

The uncertainty contribution due to job noise level sampling (value taken from Table C.4 for N = 18 and u_1 = 3,9 dB):

$c_1 u_1$ =1,4 dB

Sensitivity coefficients are:

$c_2 = c_3$ =1

The standard uncertainty, u_2, due to instrumentation is taken from Table C.5 (the instrument used was a personal sound exposure meter):

u_2 = 1,5 dB

The standard uncertainty due to the microphone position (from C.6) is:

u_3 = 1,0 dB

The combined standard uncertainty, u, of the result is derived from Formula (9):

$u^2(L_{\mathrm{EX,8h}}) = 1,4^2 + 1,5^2 + 1,0^2 = 5,2$

$u(L_{\mathrm{EX,8h}})$ = 2,3 dB

The expanded uncertainty, $U(L_{\mathrm{EX,8h}})$, is found to be (see C.3.2):

$U(L_{\mathrm{EX,8h}}) = 1,65\ u$ = 3,8 dB

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E.6.2 Final result

For an effective duration of the working day T_e = 7,5 h and for an average noise level L_{p,A,eqT_e} = 86,5 dB the daily A-weighted noise exposure level of this homogeneous noise exposure group is calculated from Formula (13):

$$L_{EX,8h} = L_{p,A,eqT_e} + 10\lg\left(\frac{T_e}{T_0}\right)\text{dB} = 86{,}2\ \text{dB}$$

The expanded uncertainty, $U(L_{EX,8h})$, is

$U(L_{EX,8h})$ = 3,8 dB

E.6.3 Conclusions

The 12 members of the homogeneous noise exposure group receive a daily A-weighted noise exposure level of 86,2 dB, with an associated expanded uncertainty for a one-sided coverage probability of 95 % (k = 1,65) of 3,8 dB.

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Annex F
(informative)

Sample calculation of daily noise exposure level using full-day measurements

F.1 General

This annex gives an example of the use of the full-day measurement strategy specified in Clause 11 in the determination of daily noise exposure level.

The example shows the application of the strategy to the task of determining the noise exposure of forklift truck drivers working in a company engaged in the manufacture and storage of wire rope.

F.2 Step 1: Work analysis

The work of the forklift truck workers involves transportation of raw materials and finished product within and between the production, storage and despatch areas of the workplace. The work of the drivers can vary depending upon the instructions of their supervisor. Trucks can be driven over a variety of surfaces while unloaded, partially loaded or fully loaded. Drivers spend a large part of the day within the truck cab, but are required to leave the cabs periodically in order to assist with loading and unloading, to discuss work with colleagues and supervisors, etc. The trucks are fitted with audible reversing alarms, the use of which is mandatory.

There are three forklift truck drivers. They work a 10 h shift, which includes three breaks of 20 min, 45 min and 20 min respectively. The two shorter breaks are taken at any convenient location within the workplace at a time to suit the driver. The longer break is taken at a fixed time in the staff canteen. The effective duration of the working day is therefore 9,25 h.

A description of the work activities was compiled by observation and confirmed by discussions with the drivers and their supervisors. The three forklift truck drivers were considered to form a homogeneous exposure group (see 7.2).

F.3 Step 2: Selection of strategy

Because the work patterns were relatively complex and unpredictable, the full-day measurement strategy was found to be most appropriate.

F.4 Step 3: Measurements

F.4.1 Measurement plan

Initially, a full-day measurement was performed on each of the drivers.

Properly calibrated personal sound exposure meters were fitted to each of the drivers at the start of the work shift. The drivers were informed as to the operation of the measuring instrument and were asked to work normally, to refrain from touching or interfering with the microphone or the measuring instrument, to try to avoid any inadvertent contact with the microphone, and to try to avoid unnecessary talking or shouting during the work shift.

The personal sound exposure meters were left undisturbed during the two shorter breaks. In this case, the noise exposure during the lunch break was considered to be irrelevant, and the personal sound exposure meters were put into “pause” mode by the measurement technician.

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The personal sound exposure meters were removed at the end of the work shift and appropriate calibration procedures were performed.

Due to the need to allow time at the beginning and end of the work shifts for setting up and removing instrumentation and instructing the drivers, the measurement duration was slightly less than the duration of the full work shift. However, the measurements were of sufficient duration to cover all significant periods of noise exposure.

Following the initial three full-day measurements, it was found that the three results differed by more than 3 dB. Therefore, three additional full-day measurements were performed, using the same techniques described above. This gave a total of six full-day measurements.

F.4.2 Observing work activities and monitoring measurements

In order to evaluate any sources of uncertainty that shall influence the results, the measurement technician periodically observed each of the drivers during the course of the measurements and made appropriate notes of their activities.

Furthermore, at the end of the work shift, the personal sound exposure meters were removed, and the technician interviewed each of the drivers to establish whether the working day was representative and to discover whether any non-typical tasks were carried out or whether any incident which would affect the measurements occurred.

F.5 Step 4: Error handling

No potential sources of error were found.

F.6 Step 5: Calculation and presentation of results and uncertainty

F.6.1 Measurement results

The results from the six measurements are shown in Table F.1.

Table F.1 — Measurement results

Driver/Day	Equivalent continuous sound pressure level $L_{p,\mathrm{A,eq}T_n}$ dB	Measurement duration
1/1	88,0	8 h 15 min
2/1	91,9	8 h 10 min
3/1	87,6	8 h 15 min
1/2	90,4	8 h 00 min
2/2	89,0	8 h 05 min
3/2	88,4	8 h 10 min

F.6.2 Calculation of daily A-weighted noise exposure level

The daily A-weighted noise exposure level of the homogeneous exposure group of forklift truck drivers is derived from the energy average of the six measured values of $L_{p,\mathrm{A,eq},T_n}$ using Formula (7).

Using the values in Table F.1, the result is $L_{p,\mathrm{A,eq},T_\mathrm{e}}$ = 89,5 dB.

The daily A-weighted noise exposure level, $L_{\mathrm{EX,8h}}$, is derived from Formula (9). The effective duration of the working day, T_e = 9,25 h and the reference duration is 8 h.

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Hence

$$L_{\mathrm{EX,8h}} = 89{,}5\,\mathrm{dB} + 10\lg\left(\frac{9{,}25}{8}\right)\mathrm{dB} = 90{,}1\,\mathrm{dB}$$

F.6.3 Calculation of uncertainty

For the full-day measurement strategy, the expanded uncertainty, U, is determined by following the procedures specified in Clause C.3.

The standard uncertainty of the energy averaged $L_{p,\mathrm{A,eq},T}$ value, u_1, is derived from Formula (C.12), i.e.

$$u_1 = \sqrt{\frac{1}{5}\left[(-1{,}2)^2 + 2{,}7^2 + (-1{,}6)^2 + 1{,}2^2 + (-0{,}2)^2 + (-0{,}8)^2\right]\mathrm{dB}} = 1{,}65\,\mathrm{dB}$$

The uncertainty contribution, $c_1 u_1$, from Table C.4 for N = 6 and u_1 = 1,65 dB is $c_1 u_1$ = 1,0 dB, which fulfils the requirement $c_1 u_1$ < 3,5 dB.

The standard uncertainty due to instrumentation, $u_{2,m}$, is taken from Table C.5, where, since the instrument used was a personal sound exposure meter:

u_2 = 1,5 dB

The standard uncertainty due to microphone position, u_3, is taken from C.6:

u_3 = 1,0 dB

The sensitivity coefficients, c_2 and c_3, are derived from Table C.3:

$c_2 = c_3 = 1$

The combined standard uncertainty, $u(L_{\mathrm{EX,8h}})$, of the result is derived from Formula (C.9):

$$u^2\left(L_{\mathrm{EX,8h}}\right) = \left(1{,}0^2 + 1{,}5^2 + 1{,}0^2\right) = 4{,}25$$

Hence, the combined standard uncertainty, $u(L_{\mathrm{EX,8h}})$ = 2,06 dB.

The expanded uncertainty, $U(L_{\mathrm{EX,8h}})$, is:

$U(L_{\mathrm{EX,8h}})$ = 1,65 u =3,4 dB

F.6.4 Conclusions

The three forklift truck drivers are subject to a daily A-weighted noise exposure level of 90,1 dB, with the associated expanded uncertainty for a one-sided coverage probability of 95 % (k = 1,65) of 3,4 dB.

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Annex G
(informative)

Example calculation of daily noise exposure level for flexible workers

G.1 General

This annex gives an example of the calculation of the daily noise exposure level for flexible workers – i.e. workers having different jobs within one day or from one day to another.

G.2 First example: flexible worker performing specific tasks in the morning and more general work during the afternoon.

In the case of a worker doing task-based work in the morning and more general work in the afternoon, the task-based approach applies.

During the morning, the worker performs the following tasks:

a) work planning;

b) cutting;

c) grinding;

d) welding.

During the afternoon the worker is performing various activities which are not related to specific tasks. The measurement is done for the entire afternoon (4 h) and then split in large samples. In this example the sample duration was 45 min in accordance with the recommendations formulated in 10.2.

The tasks measurements result and the exposure time are shown in the following table:

Table G.1

Task	Duration h	$L_{p,\mathrm{A},\mathrm{eq}T_n}$ dB
Work planning	1	70,8
Cutting	1	78,2
Grinding	1	86,2
Welding	1	85,0
Various activities	4	80,3
$L_{\mathrm{EX},8\mathrm{h}} = 81{,}9\,\mathrm{dB}$		

The daily exposure is calculated as indicated in 9.5 and the associated uncertainty is calculated according to C.2.

G.3 Second example: flexible worker performing different jobs from day to day

The example shows how to calculate the daily noise exposure for a flexible worker whose jobs for a 5 day period are distributed as follows.

— 1 day as a forklift worker (the full day strategy being the most suitable approach);

— 2,5 days as an worker on a production line (the job-based strategy being the most suitable approach);

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— 1,5 days as an worker on a specific machine (the task-based strategy being the most suitable approach).

G.3.1 Noise exposure calculation as a forklift worker

Measurements made according to the "full day measurement" strategy:

Table G.2

Equivalent continuous sound pressure level $L_{p,\mathrm{A,eq}T_n}$ dB		
Measurement 1	Measurement 2	Measurement 3
83,5	82	81,8

In accordance with 11.3, the average noise level, $L_{p,\mathrm{A,eq}T_n}$, is calculated from Formula (7).

$$L_{p,\mathrm{A,eq}T_e} = 82{,}5\,\mathrm{dB} \text{ with } U = 3{,}8\,\mathrm{dB}$$

For an effective duration of the working day T_e = 7,5 h and for an average noise level $L_{p,\mathrm{A,eq}T}$ = 82,5 dB, the daily A-weighted noise exposure level is calculated from Formula (9):

$$L_{\mathrm{EX,8h}} = 82{,}2\,\mathrm{dB} \text{ and } U = 3{,}8\,\mathrm{dB}$$

G.3.2 Noise exposure calculation as a worker on a production line

Measurements made according to the "job-based measurements" strategy:

Table G.3

Equivalent continuous sound pressure level $L_{p,\mathrm{A,eq}T_n}$ dB				
Measurement 1	Measurement 2	Measurement 3	Measurement 4	Measurement 5
85,6	84,3	85,2	83,8	85,1

In accordance with 10.4, the average noise level, $L_{p,\mathrm{A,eq}T_\mathrm{e}}$, is calculated from Formula (7).

$$L_{p,\mathrm{A,eq}T_\mathrm{e}} = 84{,}8\,\mathrm{dB} \text{ and } U = 3{,}1\,\mathrm{dB}$$

For an effective duration of the working day T_e = 7,5 h and for an average noise level $L_{p,\mathrm{A,eq}T_\mathrm{e}}$ = 84,8 dB, the daily A-weighted noise exposure level is calculated from Formula (9):

$$L_{EX,8\mathrm{h}} = 84{,}6\,\mathrm{dB} \text{ and } U = 3{,}1\,\mathrm{dB}$$

G.3.3 Noise exposure calculation as a worker on a specific machine

Measurements made according to the "task-based measurements" strategy:

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Table G.4

Equivalent continuous sound pressure level $L_{p,\mathrm{A,eq},T_n}$ dB				
Task	**Duration** h	**Measurement 1**	**Measurement 2**	**Measurement 3**
Task 1	0,5	70,8	71,2	69,4
Task 2	3	78,2	80,1	79,7
Task 3	4	86,2	85,8	85,2

For each task, the average noise level is calculated using Formula (7) and the daily A-weighted noise exposure level is calculated according to Formula (8).

Task 1: $L_{p,\mathrm{A,eq}T_1} = 70{,}5\,\mathrm{dB}$ and $L_{\mathrm{EX,8h},1} = 58{,}5\,\mathrm{dB}$

Task 2: $L_{p,\mathrm{A,eq}T_2} = 79{,}4\,\mathrm{dB}$ and $L_{\mathrm{EX,8h},2} = 75{,}1\,\mathrm{dB}$

Task 3: $L_{p,\mathrm{A,eq}T_3} = 85{,}8\,\mathrm{dB}$ and $L_{\mathrm{EX,8h},3} = 82{,}8\,\mathrm{dB}$

The daily A-weighted noise exposure level is then calculated from Formula (6):

$L_{\mathrm{EX,8h}} = 83{,}5\,\mathrm{dB}$ and $U = 2{,}6\,\mathrm{dB}$

G.4 Combined daily noise exposure of the flexible worker

The combined daily A-weighted noise exposure is calculated according to formula in 3.2:

$$L_{\mathrm{EX,8h}} = 10\lg\left[\left(1\times10^{0,1\times82,2} + 2{,}5\times10^{0,1\times84,6} + 1{,}5\times10^{0,1\times83,5}\right)/5\right] = 83{,}9\,\mathrm{dB}$$

The standard uncertainty due to instrumentation, $u_{2,m}$, is taken from Table C.5, where, since the instrument used was a personal sound exposure meter:

u_2 = 1,5 dB

The standard uncertainty due to microphone position, is taken from C.6:

u_3 = 1 dB

The associated uncertainty is calculated according to C.7Formula (C.14):

$$\bar{U}^2 = \left[\frac{1\times10^{0,1\times82,2}}{\left(1\times10^{0,1\times82,2} + 2{,}5\times10^{0,1\times84,6} + 1{,}5\times10^{0,1\times83,5}\right)}\right]^2 \times 3{,}8^2$$
$$+\left[\frac{2{,}5\times10^{0,1\times84,6}}{\left(1\times10^{0,1\times82,2} + 2{,}5\times10^{0,1\times84,6} + 1{,}5\times10^{0,1\times83,5}\right)}\right]^2 \times 3{,}1^2$$

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$$+\left[\frac{1{,}5\times10^{0{,}1\times83{,}5}}{\left(1\times10^{0{,}1\times82{,}2}+2{,}5\times10^{0{,}1\times84{,}6}+1{,}5\times10^{0{,}1\times83{,}5}\right)}\right]^2\times2{,}6^2=4{,}1$$

The final daily noise exposure is:

$$L_{\mathrm{EX,8h}}=83{,}9\,\mathrm{dB}\ \text{with}\ U=2{,}0\,\mathrm{dB}$$

The spreadsheet in https://standards.iso.org/iso/9612/ed-3/en allows users to calculate exposure and uncertainty with that method.

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Annex H
(normative)

Uncertainty calculation for peak sound pressure levels

H.1 Forewarning

This document requires the evaluation of the $L_{p,\mathrm{Cpeak}}$ where significant noise events can occur, resulting in a change of the noise level or which can result in peak levels exceeding regional regulation limits (see 7.2).

Identification of the noise sources and tasks that leads to these significant noise events is essential for a correct determination of the highest peak levels $L_{p,\mathrm{Cpeak}}$ (see 9.1).

Different sources of such high-noise isolated events exist, e.g.:

a) some external activity (e.g. shooting at a close range, use of heavy weapons),

b) a part of the overall manufacturing process which is not directly related to the specific workplace (e.g. steam release, pneumatic exhaust, track firecracker),

c) a part of the work task (hammering, use of explosives).

When two or more competing families of high-noise events occur (in terms of possible maximum among peak levels), it is recommended to undertake the evaluation for each family independently and to report results for all families.

If unexplained peak levels occur, the influence on the result shall be investigated, and the measurement shall be repeated if the result is significantly affected (see 13.2).

It can be assumed that each event of the formed family is a random event, and its peak level is a random variable. In their physical nature, peak levels are bounded from above. There are two possible scenarios:

— only one event has been observed, and it is fortuitous and therefore essentially unique,

— the event is repeated, and the upper limit of the peak sound level for this event is to be estimated from sampling, i.e. the $L_{p,\mathrm{Cpeak}}$ is the upper limit of the finite support for the random variable, and the estimate of the $L_{p,\mathrm{Cpeak}}$ requires to evaluate that upper limit on the basis of a given sampling of occurrences of the event.

H.2 Unique event

If the significant noise events where the maximum peak levels were observed are entirely sampled, the $L_{p,\mathrm{Cpeak}}$ maximum is simply the largest of the measured values. Therefore, the associated uncertainty can be assigned to the instrumentation and the position of the microphone by Formula (H.1):

$$u = \sqrt{c_2^2 u_2^2 + c_3^2 u_3^2} \tag{H.1}$$

The instrumentation uncertainty u_2 is likely larger than the value of 0,7 dB which applies to the uncertainty on $L_{p,\mathrm{A,eq}T}$. Values as large as 1,4 to 2,4 dB are possible (see IEC 61672-1, Table 4, 5.12.3). A value u_2 = 1,4 dB is recommended, which is twice as large as the value used for the equivalent sound pressure level.

The instrumentation position uncertainty remains the same than for $L_{p,\mathrm{A,eq}T}$, see (C.6.). However, high-level peaks are often generated by high energy impacts between two rigid objects. These impacts have generally a large high-frequency content, and as such they can be very directional. Additionally, microphones

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themselves are very directional at high frequency. In such cases, the instrumentation position uncertainty can be larger than the estimate for continuous noises. It is then recommended to experimentally evaluate it, insofar as its contribution will be important compared to that of u_1 and u_2.

H.3 Finite sampling

If the uncertainty of measurement associated with the peak sound level is evaluated based on sampling, it is necessary to know the nature of the distribution to which these samples conform. There are two options:

a) if a sufficient number of measurements are available, it is possible to run a statistical test to find out which distribution law they comply with, and to check their statistical significance;

b) otherwise, a assumption has to be made, for example, based on what is reported in the literature.

When the distribution law of the samples is known, it becomes possible to estimate the uncertainty on the peak sound level. An example is given for uniform distribution, the most widely encountered. For other distribution laws, please refer to the literature (see in particular references below) or to the GUM.

H.3.1 Case of a uniform distribution

Provided that there exist lower and upper bounds for $L_{p,\mathrm{Cpeak}}$, it can be considered as a random variable ξ distributed within those bounds. In the absence of other information, it is reasonable, according to the GUM recommendation, to assign it the uniform distribution $U_\xi\,(l-\Delta;l)$ where Δ is the unknown support, and l is the sought value.

Assuming that n measurements of the peak level $\{y_i\}_{i=1:n}$ have been carried out, the posterior distribution for $\{\Delta,l\}, W\{\Delta,l\}$, can be derived from the Bayes' theorem. Since Δ and l are the location and scale parameter accordingly, the Jeffreys prior is Δ^{-1} and the likelihood function is $\prod_{i=1}^{n} U_\xi\,(l-\Delta;l)$. Thus, according to the Bayes' theorem given by Formula (H.2):

$$W(\Delta,l\,|\,y_1,\ldots,y_n)\propto\Delta^{-1}\prod_{i=1}^{n}U_\xi\,(l-\Delta;l)=\Delta^{-(n+1)}\,f(\Delta,l\,|\,y_{\min},y_{\max}) \tag{H.2}$$

Where

$$f(\Delta,l\,|\,y_{\min},y_{\max})=\begin{cases}1;\,l\in[y_{\min},y_{\max}+\Delta]\\0;\,l\notin[y_{\min},y_{\max}+\Delta]\end{cases}$$

$$y_{\min}=\min\{y_i\},\,y_{\max}=\max\{y_i\}$$

It follows from the Formula (H.2) that $W(\Delta,l\,|\,y_1,\ldots,y_n)$ is proportional to $\Delta^{-(n+1)}$ and defined over the region bounded by the straight line $l=y_{\max}$ from below and by the straight line $l=\Delta+y_{\min}$ from above; $l\geq y_{\max}$, $\Delta\geq y_{\max}-y_{\min}$.

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Posterior distribution for the upper bound $l, W(l)$, can be obtained by integrating $W(\Delta, l \mid y_1, \ldots, y_n)$ over Δ, that yields to Formula (H.3):

$$W(l) \propto \frac{1}{n(l - y_{\min})^n}, l \geq y_{\max} \tag{H.3}$$

Normalization leads to the following final Formula (H.4) for $W(l)$:

$$W(l) = \frac{(n-1)(y_{\max} - y_{\min})^{n-1}}{(l - y_{\min})^n} l \geq y_{\max} \tag{H.4}$$

Its mathematical expectation is expressed in Formula (H.5):

$$\mu = y_{\max} + \frac{(y_{\max} - y_{\min})}{n-2} \tag{H.5}$$

And variance σ^2 can be calculated by Formula (H.6):

$$\sigma^2 = (n-1)(y_{\max} - y_{\min})^{n-1} \int_{y_{\max}}^{\infty} \frac{\left[(l - y_{\max}) - \frac{y_{\max} - y_{\min}}{n-2}\right]^2}{(l - y_{\min})^n} \mathrm{d}l \tag{H.6}$$

Which yields to the expression of the standard deviation Formula (H.7):

$$\sigma = (y_{\max} - y_{\min}) \sqrt{\frac{n-1}{(n-2)^2 (n-3)}}, n \geq 4 \tag{H.7}$$

Final result $L_{p,\mathrm{Cpeak},95\,\%}$:

The best estimate of $L_{p,\mathrm{Cpeak}}$, noted $\breve{L}_{p,\mathrm{Cpeak}}$, for a sampling of n measurement $\left\{\breve{L}_{p,\mathrm{Cpeak,min}}, \ldots, \breve{L}_{p,\mathrm{Cpeak,max}}\right\}$ is given in Formula (H.8):

$$\breve{L}_{p,\mathrm{Cpeak}} = \breve{L}_{p,\mathrm{Cpeak,max}} + \frac{\left(\breve{L}_{p,\mathrm{Cpeak,max}} - \breve{L}_{p,\mathrm{Cpeak,min}}\right)}{n-2} \tag{H.8}$$

And the associated standard uncertainty is given in Formula (H.9):

$$u_1\left(\breve{L}_{p,\mathrm{Cpeak}}\right) = \left(\breve{L}_{p,\mathrm{Cpeak,max}} - \breve{L}_{p,\mathrm{Cpeak,min}}\right) \sqrt{\frac{n-1}{(n-2)^2 (n-3)}}, n \geq 4 \tag{H.9}$$

For the purpose of this document, one-sided confidence interval of 95 % is considered, which results in an enlargement factor k = 2 (uniform law). This means that 95 % of the estimated values are above the real upper limit given by Formula (H.10):

$$L_{p,\mathrm{Cpeak},95\,\%} = \breve{L}_{p,\mathrm{Cpeak}} + 2 \times u_1\left(\breve{L}_{p,\mathrm{Cpeak}}\right) \tag{H.10}$$

NOTE The question of estimate robustness always arises, especially when n is small. For example, if the random variables are mainly grouped near the lower bound, the $L_{p,\mathrm{Cpeak}}$ can be grossly overestimated or grossly underestimated depending on the proximity of $\breve{L}_{p,\mathrm{Cpeak,max}}$ to the true upper bound. Two typical approaches can be used to make the estimate more robust:

— Use an order statistic as the estimator. In this case it means to take $\breve{L}_{p,\mathrm{Cpeak}} = \breve{L}_{p,\mathrm{Cpeak,max}}$ and the standard uncertainty as the average squared deviation from the estimate:

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$$u_1\left(\breve{L}_{p,\mathrm{Cpeak}}\right)=\left(\breve{L}_{p,\mathrm{Cpeak,max}}-\breve{L}_{p,\mathrm{Cpeak,min}}\right)\sqrt{\frac{2}{(n-2)(n-3)}},n\geq 4 \quad \text{(H.11)}$$

Thus, the underestimation of the sought upper bound is “compensated” by a higher standard uncertainty.

— Elimination of “outliers”. This means that only samples of the highest orders are to be left in the sample. That can result in the following possible procedure:

a) obtain a sample of size n and retain only the five (5) highest values;

b) use the estimate $\breve{L}_{p,\mathrm{Cpeak}}=\breve{L}_{p,\mathrm{Cpeak,max}}$ with the relevant standard uncertainty as calculated for the truncated sample, n = 5 (see Formula (H.11)).

H.3.2 Other approaches

Reference [13] proposes an approach for cases where the assumption of a normal sample distribution cannot be rejected.

References [14] and [15] proposes using the statistical law of extreme value, a priori well suited when a maximum value is sought from a set of samples.

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